

BAUM TenPers Bitcoin Analysis

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1 Computational Framework

1.1 program_data.xlsx

Date	SPXT (NLC)	BTC_USD (NLC)	BCOM (C)	USD (C)	VIX (C)	US10Y (C)	US1M (C)
2020-01-01	6553.57	8964.53	80.89	114.67	13.78	1.9175	1.4299
2020-01-02	6609.29	8705.95	81.00	114.97	12.47	1.8771	1.4604
2020-01-03	6568.32	9081.59	81.01	114.99	14.02	1.8781	1.4886
2020-01-06	6563.32	9081.59	81.37	114.99	14.02	1.7881	1.4886
2020-01-07	6586.54	9467.66	81.64	114.96	13.85	1.8089	1.5089
2020-01-08	6601.15	10000.76	80.58	115.13	13.45	1.8738	1.4961

Table 1: Snippet of US Regression sheet (short schema). NLC = $\Delta \log(\cdot)$, C = first difference.

Date	IDX (NLC)	BTC_LCY (NLC)	BCOM (C)	USD (C)	VIX (C)	US10Y (C)	US1M (C)	LOC10Ysp (C)	LOC3Msp (C)
2020-01-01	7542.40	6763.61	80.89	114.67	13.78	1.9175	1.4299	-2.324	0.6840
2020-01-02	7604.30	6623.03	81.00	114.97	12.47	1.8771	1.4604	-2.424	0.7040
2020-01-03	7622.40	6241.52	81.01	114.99	14.02	1.7881	1.4886	-2.496	0.6880
2020-01-06	7622.40	6941.65	81.37	114.99	14.02	1.7881	1.4886	-2.496	0.6880
2020-01-07	7575.30	7188.26	81.64	114.96	13.85	1.8089	1.5089	-2.486	0.7150
2020-01-08	7575.80	10033.88	80.58	115.13	13.79	1.8777	1.4885	-2.477	0.7190

Table 2: Representative international sheet. The exact same column order applies to UK Regression, Eurozone Regression, Japan Regression, China Regression, Brazil Regression, India Regression, and South Africa Regression; only the tickers change on a country specific basis.

Data feed for regression.py This workbook is the direct input to the regression program: the script reads the sheets, applies the suffix-based transforms, and produces the $\text{\LaTeX}$ tables. Both the workbook and the regression run are *coin-specific*. The examples shown use Bitcoin series (e.g., BTC_USD, BTC_GBP), but identical tables for other currencies—such as Ripple/XRP or Litecoin/LTC—are obtained by following the same template and supplying a workbook of the same shape in which the BTC columns are replaced by the corresponding coin columns (e.g., BTC_USD $\rightarrow$ XRP_USD or LTC_USD, BTC_GBP $\rightarrow$ XRP_GBP or LTC_GBP, etc.). No other formatting changes are required: the sheet order, headers, and suffixes (NLC, C) remain the same. Running the program against that coin-specific `program_data.xlsx` generates the matching coin-specific regression tables.

Data feed for wedge_var_ratio.py Duplicate the currency-specific workbook and delete every sheet except the target country tab so one sheet remains. For a *country-specific* change, edit the quoted names in the program so the equity header matches the index used on that country's sheet—e.g., switch to "UKX Index" when using the UK Regression tab, to "SXXP Index" on Eurozone Regression, to "NKYTR Index" on Japan Regression, etc. If that country's sheet includes local spread controls, add those headers to CTRL by editing the quoted entries—e.g., the UK sheet uses "GTGBP10YR @BGN Corp" and "GTGBP3MO @BGN Corp"; the Eurozone sheet uses "GTEUR10YR @BGN Corp" and "GTEUR3MO @BGN Corp"; Japan uses "GTJPY10YR @BGN Corp" and "GTJPY3MO @BGN Corp", etc. For a *coin-specific* change, keep the country

headers fixed and edit only the quoted coin column so BTC points to the coin used on that same sheet—e.g., on the US sheet change "BTC_USD" to "XRP_USD" or "LTC_USD"; on the UK sheet change "BTC_GBP" to "XRP_GBP" or "LTC_GBP"; on the Eurozone sheet change "BTC_EUR" to "XRP_EUR" or "LTC_EUR"; similarly for JPY, CNY, BRL, INR, and ZAR on their respective sheets. Run the script on this one-sheet workbook; it will read that sheet, apply the suffix-based transforms, and compute the wedge variance ratio for that specific country-coin pair.

1.2 regression.py

```
import pandas as pd
import numpy as np
import statsmodels.api as sm

# ===== INPUT WORKBOOK (SHEETS HOLD **PRICE LEVELS**) =====
file = "program_data.xlsx"
xls = pd.ExcelFile(file)

# ===== SPECS (TRANSFORM IS DICTATED BY THE SUFFIX) =====
regression_specs = {
    "US Regression": {
        "dependent": "SPXT Index (Natural Log Changes)",
        "independents": [
            "BTC_USD (Natural Log Changes)",
            "BCOM Index (Changes)",
            "USTWBGD Index (Changes)",
            "VIX Index (Changes)",
            "USGG10YR Index (Changes)",
            "USGG1M Index (Changes)"
        ]
    },
    "UK Regression": {
        "dependent": "UKX Index (Natural Log Changes)",
        "independents": [
            "BTC_GBP (Natural Log Changes)",
            "BCOM Index (Changes)",
            "USTWBGD Index (Changes)",
            "VIX Index (Changes)",
            "USGG10YR Index (Changes)",
            "USGG1M Index (Changes)",
            "GTGPPII10YR @BGN Corp (Changes)",
            "GTGP3MO @BGN Corp (Changes)"
        ]
    },
    "Eurozone Regression": {
        "dependent": "SXXP Index (Natural Log Changes)",
        "independents": [
            "BTC_EUR (Natural Log Changes)",
            "BCOM Index (Changes)",
            "USTWBGD Index (Changes)",
            "VIX Index (Changes)",
            "USGG10YR Index (Changes)",
            "USGG1M Index (Changes)",
            "GTEUR10YR @BGN Corp (Changes)",
            "GTEUR3MO @BGN Corp (Changes)"
        ]
    },
    "Japan Regression": {
        "dependent": "NKYTR Index (Natural Log Changes)",
        "independents": [
            "BTC_JPY (Natural Log Changes)",
            "BCOM Index (Changes)",
            "USTWBGD Index (Changes)",
            "VIX Index (Changes)",
            "USGG10YR Index (Changes)",
            "USGG1M Index (Changes)",
            "GTJPY10YR @BGN Corp (Changes)",
            "GTJPY3MO @BGN Corp (Changes)"
        ]
    },
    "China Regression": {
        "dependent": "SHCOMP Index (Natural Log Changes)",
        "independents": [
            "BTC_CNY (Natural Log Changes)",
            "BCOM Index (Changes)",
            "USTWBGD Index (Changes)",
            "VIX Index (Changes)"
        ]
    }
}
```

```

        USGG10YR Index (Changes) ,
        USGG1M Index (Changes) ,
        GTCNY10YR @CHBE Corp (Changes) ,
        GTCNY1YR @CHBE Corp (Changes)
    ]
},
Brazil Regression : {
    dependent : IBOV Index (Natural Log Changes) ,
    independents : [
        BTC_BRL (Natural Log Changes) ,
        BCOM Index (Changes) ,
        USTWBGD Index (Changes) ,
        VIX Index (Changes) ,
        USGG10YR Index (Changes) ,
        USGG1M Index (Changes) ,
        GTBRL10YR @BGN Corp (Changes) ,
        GTBRL1YR @BGN Corp (Changes)
    ]
},
India Regression : {
    dependent : NIFTY Index (Natural Log Changes) ,
    independents : [
        BTC_INR (Natural Log Changes) ,
        BCOM Index (Changes) ,
        USTWBGD Index (Changes) ,
        VIX Index (Changes) ,
        USGG10YR Index (Changes) ,
        USGG1M Index (Changes) ,
        GTINR10YR @NDSI Corp (Changes) ,
        GTINR2YR @NDSI Corp (Changes)
    ]
},
South Africa Regression : {
    dependent : JALSH Index (Natural Log Changes) ,
    independents : [
        BTC_ZAR (Natural Log Changes) ,
        BCOM Index (Changes) ,
        USTWBGD Index (Changes) ,
        VIX Index (Changes) ,
        USGG10YR Index (Changes) ,
        USGG1M Index (Changes) ,
        GTZAR10YR @BGN Corp (Changes) ,
        GTZAR1YR @BGN Corp (Changes)
    ]
}
}

# ===== LABELS (FOR THE TABLE) =====
label_map = {
    BTC_USD (Natural Log Changes) : r  $\Delta \log(\mathrm{BTC}_{\{\mathrm{USD}\}})$  ,
    BTC_GBP (Natural Log Changes) : r  $\Delta \log(\mathrm{BTC}_{\{\mathrm{GBP}\}})$  ,
    BTC_EUR (Natural Log Changes) : r  $\Delta \log(\mathrm{BTC}_{\{\mathrm{EUR}\}})$  ,
    BTC_JPY (Natural Log Changes) : r  $\Delta \log(\mathrm{BTC}_{\{\mathrm{JPY}\}})$  ,
    BTC_CNY (Natural Log Changes) : r  $\Delta \log(\mathrm{BTC}_{\{\mathrm{CNY}\}})$  ,
    BTC_BRL (Natural Log Changes) : r  $\Delta \log(\mathrm{BTC}_{\{\mathrm{BRL}\}})$  ,
    BTC_INR (Natural Log Changes) : r  $\Delta \log(\mathrm{BTC}_{\{\mathrm{INR}\}})$  ,
    BTC_ZAR (Natural Log Changes) : r  $\Delta \log(\mathrm{BTC}_{\{\mathrm{ZAR}\}})$  ,
    BCOM Index (Changes) : r  $\Delta \mathrm{Commodity}$  ,
    USTWBGD Index (Changes) : r  $\Delta \mathrm{Dollar}$  ,
    VIX Index (Changes) : r  $\Delta \mathrm{VIX}$  ,
    USGG10YR Index (Changes) : r  $\Delta \mathrm{10Y\ Yield}$  ,
    USGG1M Index (Changes) : r  $\Delta \mathrm{1M\ Yield}$  ,
    GTBRL10YR @BGN Corp (Changes) : r  $\Delta \mathrm{10Y\ Spread}$  ,
    GTBRL1YR @BGN Corp (Changes) : r  $\Delta \mathrm{1Y\ Spread}$  ,
    GTGBP10YR @BGN Corp (Changes) : r  $\Delta \mathrm{10Y\ Spread}$  ,
    GTGBP3MO @BGN Corp (Changes) : r  $\Delta \mathrm{3M\ Spread}$  ,
    GTEUR10YR @BGN Corp (Changes) : r  $\Delta \mathrm{10Y\ Spread}$  ,
    GTEUR3MO @BGN Corp (Changes) : r  $\Delta \mathrm{3M\ Spread}$  ,
    GTJPY10YR @BGN Corp (Changes) : r  $\Delta \mathrm{10Y\ Spread}$  ,
    GTJPY3MO @BGN Corp (Changes) : r  $\Delta \mathrm{3M\ Spread}$  ,
    GTCNY10YR @CHBE Corp (Changes) : r  $\Delta \mathrm{10Y\ Spread}$  ,
    GTCNY1YR @CHBE Corp (Changes) : r  $\Delta \mathrm{1Y\ Spread}$  ,
    GTINR10YR @NDSI Corp (Changes) : r  $\Delta \mathrm{10Y\ Spread}$  ,
    GTINR2YR @NDSI Corp (Changes) : r  $\Delta \mathrm{2Y\ Spread}$  ,
    GTZAR10YR @BGN Corp (Changes) : r  $\Delta \mathrm{10Y\ Spread}$  ,
    GTZAR1YR @BGN Corp (Changes) : r  $\Delta \mathrm{1Y\ Spread}$ 
}

```

This code snippet declares which columns are the dependent and independent variables for each country, how those columns must be transformed, and how each variable should be rendered in $\text{\LaTeX}$. The inputs in

the workbook are *level* series.

Top controls. The line `FILE="program_data.xlsx"` instructs the script to open that workbook (one sheet per country). `xls = pd.ExcelFile(FILE)` prepares a handle; each loop then loads a specific sheet according to its name. Note, date filtering is not performed by the script; any sample windowing must be handled by the user directly in the Excel workbook by trimming rows on each country specific sheet.

How transformations are encoded. The preprocessing rule is encoded *only* by the column-name suffix. Any header that ends with *(Natural Log Changes)* is converted to $\Delta \log(\cdot)$; any header that ends with *(Changes)* is turned into a first difference $\Delta(\cdot)$. The helper `transform_from_levels` applies this mapping and returns a Series whose *name exactly matches the input header*. Headers lacking one of these two suffixes raise an error.

What regression_specs contains. `regression_specs` is a Python dictionary with one entry per region (sheet). Each entry has three keys. The value of `dependent` is the equity index header with the *(Natural Log Changes)* suffix, so the left-hand side is $y_t = \Delta \log(\text{equity})$. The value of `independents` is an ordered list whose first element is Bitcoin in the local currency, again with the *(Natural Log Changes)* suffix, so the main regressor is $\Delta \log(\text{BTC}_{\text{local}})$. The remaining elements are macro and rate controls with the *(Changes)* suffix; the script differences those levels to create ΔBCOM , $\Delta \text{USTWBGD}$, ΔVIX , and the yield/spread deltas.

What is common across countries. Every country uses the same core control set, all in level differences: `BCOM` (commodities), `USTWBGD` (broad dollar), `VIX`, and two U.S. rate deltas (`USGG10YR` and `USGG1M`). These give the script five lines in the list that always mean “difference this level series.” On top of these, each country appends two local term-structure series, also marked with *(Changes)* so they are differenced. The equity on the left and the BTC series on the right are always in natural-log changes and always appear as the first two declarations in each block.

What differs by country (explained once, not repeated). The US block sets the dependent to *SPXT Index (Natural Log Changes)* and uses *BTC_USD (Natural Log Changes)*; it relies only on the common controls. The UK block uses *UKX* and *BTC_GBP* and appends *GTGBPII10YR @BGN Corp (Changes)* and *GTGBP3MO @BGN Corp (Changes)*. The Eurozone block uses *SXXP* and *BTC_EUR* and appends *GTEUR10YR @BGN Corp (Changes)* and *GTEUR3MO @BGN Corp (Changes)*. Japan uses *NKYTR* and *BTC_JPY* and appends *GTJPY10YR @BGN Corp (Changes)* and *GTJPY3MO @BGN Corp (Changes)*. China uses *SHCOMP* and *BTC_CNY* and appends *GTCNY10YR @CHBE Corp (Changes)* and *GTCNY1YR @CHBE Corp (Changes)*. Brazil uses *IBOV* and *BTC_BRL* and appends *GTBRL10YR @BGN Corp (Changes)* and *GTBRL1YR @BGN Corp (Changes)*. India uses *NIFTY* and *BTC_INR* and appends *GTINR10YR @NDSI Corp (Changes)* and *GTINR2YR @NDSI Corp (Changes)*. South Africa uses *JALSH* and *BTC_ZAR* and appends *GTZAR10YR @BGN Corp (Changes)* and *GTZAR1YR @BGN Corp (Changes)*. In every case the `currency` line mirrors the BTC quote and functions only as a selector tag for the caption.

How the labels are produced. After the country specifications, `label_map` converts raw headers into L^AT_EX-ready math strings using raw-string literals to preserve backslashes. Thus *BTC_USD (Natural Log Changes)* prints as $\Delta \log(\text{BTC}_{\text{USD}})$, *BCOM Index (Changes)* as $\Delta \text{Commodity}$, and each local tenor as a concise object such as $\Delta 10\text{Y Spread}$ or $\Delta 3\text{M Spread}$. The mapping is for presentation only; it does not affect data.

What the runner will actually build. For a chosen country sheet, the script reads `FILE`, transforms the dependent as $\Delta \log(\text{equity})$, transforms BTC as $\Delta \log(\text{BTC}_{\text{local}})$, differences every control marked *(Changes)*, and then *row-aligns* all series via an inner join and `dropna()`. It fits a *stepwise* sequence of OLS models with an explicit intercept: column (1) includes only BTC, column (2) adds the next control, and so on until all regressors are included. Coefficients are printed with conventional significance stars (`***`, `**`, `*`), standard errors are shown in parentheses on the next line, and the footer reports Observations and adjusted R^2 .

```

def stars(p):
    return *** if p < 0.01 else ( ** if p < 0.05 else ( * if p < 0.1 else  ))

# ===== CORE: BUILD SERIES FROM **LEVEL** COLUMNS =====
def from_levels(df: pd.DataFrame, header_with_suffix: str) -> pd.Series:
    if header_with_suffix.endswith( (Natural Log Changes) ):
        base = header_with_suffix[:-len( (Natural Log Changes) )]
        s = df[base].astype(float)
        out = np.log(s).diff()
    elif header_with_suffix.endswith( (Changes) ):
        base = header_with_suffix[:-len( (Changes) )]
        s = df[base].astype(float)
        out = s.diff()
    else:
        raise ValueError(f Unknown suffix in header: {header_with_suffix} )
    return out.rename(header_with_suffix)

```

stars(p). The function returns a short marker based on the input p-value. If $p < 0.01$ it returns "***", if $p < 0.05$ it returns "**", if $p < 0.1$ it returns "*", and otherwise it returns the empty string. The intended usage is to append this marker directly to a formatted coefficient, e.g. `f"{coef:.3f}{stars(p)}"` so that the star logic remains isolated from the printing logic.

from_levels(df, header_with_suffix). The input `header_with_suffix` is a column name that encodes the desired preprocessing rule in its suffix. The function first checks whether the header ends with (Natural Log Changes) or with (Changes). It then strips that suffix to recover the base header, pulls the corresponding level series from `df`, coerces it to floating point, and applies the appropriate transform. For (Natural Log Changes) it computes $\Delta \log(\text{level})$ via `np.log(s).diff()`, and for (Changes) it computes the first difference $\Delta(\text{level})$ via `s.diff()`. Headers that do not end with one of these two exact suffixes trigger a `ValueError` that includes the offending header for fast debugging.

Name contract and alignment. After transforming, the function renames the returned `Series` back to the full input header, including the suffix. This preserves a one-to-one contract between the name a caller passes in and the name used everywhere downstream for model assembly and table printing. Because differencing introduces a leading NaN, the caller typically concatenates all transformed columns and calls `dropna()` once to define the common sample window.

```

# ===== RUN ALL SHEETS, FIT NESTED MODELS, WRITE ONE TEX =====
all_lines = []

for sheet, spec in regression_specs.items():
    df = pd.read_excel(xls, sheet_name=sheet)

    # Build y and every X directly from LEVEL columns in the sheet
    y_full = from_levels(df, spec[ dependent ])
    X_all = {name: from_levels(df, name) for name in spec[ independents ]}
    X_all = pd.DataFrame(X_all)

    models = []
    order = spec[ independents ]
    for i in range(1, len(order) + 1):
        T = pd.concat([y_full, X_all[order[:i]]], axis=1).dropna()
        y = T[spec[ dependent ]]
        X = sm.add_constant(T[order[:i]], has_constant= add )
        models.append(sm.OLS(y, X).fit())

    # LaTeX table
    k = len(models)
    lines = []
    lines.append( \begin{table}[H] )
    lines.append( \centering )
    lines.append( \caption{TODO} )
    lines.append( \resizebox{\textwidth}{!}{% )
    lines.append( \begin{tabular}{l + c * k + } )
    lines.append( \toprule )
    lines.append( & + & .join(f ({j+1}) for j in range(k)) + \\\ )
    lines.append( \midrule )

```

```

for var in order:
    label = label_map.get(var, var)
    coef_row, se_row = [], []
    for m in models:
        if var in m.params.index:
            coef_row.append(f {m.params[var]:.3f}{stars(m.pvalues[var])} )
            se_row.append(f ({m.bse[var]:.3f}) )
        else:
            coef_row.append( )
            se_row.append( )
    lines.append(f {label} & + & .join(coef_row) + \\\ )
    lines.append( & + & .join(se_row) + \\\ )

lines.append( \\midrule )
lines.append( Observations & + & .join(f {int(m.nobs)} for m in models) + \\\ )
lines.append( Adj. $R^2$ & + & .join(f {m.rsquared_adj:.3f} for m in models) + \\\ )
lines.append( \\bottomrule )
lines.append( \\end{tabular} )
lines.append( } ) # close resizebox
lines.append( \\begin{tablenotes} )
lines.append( \\footnotesize )
lines.append( Note: Dependent variable shown in caption. SE in parentheses. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$. )
lines.append( \\end{tablenotes} )
lines.append( \\end{table}\\n )

all_lines.extend(lines)

with open( regressions.tex , w ) as f:
    f.write( \\n .join(all_lines))

```

Loop and sheet loading. The code iterates over the specification dictionary and, for each entry, reads the Excel tab whose name equals the spec key. This establishes a one-to-one mapping between a spec and its sheet without relying on sheet order.

Building transformed series. Within a sheet, the script constructs the dependent series y and every regressor X_j directly from the *level* columns by calling `from_levels`. The function interprets the header suffix to apply either $\Delta \log(\cdot)$ or first differences, and it returns a Series whose name matches the header with suffix. The list `order` preserves the declared regressor order for the stepwise models and for printing.

Stepwise model fitting. For $i = 1, \dots, K$ (where K is the number of regressors in `order`), the code forms a temporary data set T by concatenating y and the first i regressors, then calls `dropna()` once. This single operation both removes the leading undefined value created by differencing and enforces a common sample across the variables included in model i . The design matrix is then $\alpha + X_{1:i}$ using `sm.add_constant(..., has_constant="add")`. Each fitted OLS model is appended to `models` so the table can print columns (1), ..., (K).

Coefficient and standard-error extraction. For each regressor in `order`, the code scans across the fitted models and, when the parameter is present in a given model, pulls the point estimate, attaches significance marks via `stars(p)`, and stores the corresponding standard error. If a variable is not in an early column (because it has not yet entered the stepwise set), that cell is left blank in both the coefficient row and the parentheses row.

Table assembly. The table body is created programmatically. The header line lists (1) through (K). Each regressor contributes two printed lines: the coefficient row with stars and the next line with its standard errors in parentheses. The footer reports the number of observations n for each column and the adjusted R^2 . A `resizebox` encloses the `tabular` so wide panels fit the page. The caption placeholder is set to `TODO` in the snippet; it can be replaced with any text (for example, the country name) without changing data handling.

Labels and raw names. Displayed variable names come from `label_map` when an entry exists; otherwise the raw header is printed. Because `from_levels` returns series whose names equal the headers with suffixes, indexing the model output by those exact strings is consistent during extraction and printing.

Output and order. All L^AT_EX lines for the current sheet are appended to `all_lines`. After the loop completes, the script writes a single file `regressions.tex` that contains the tables for every processed sheet in the same order as the dictionary iteration.

What the sample means for each column. Since `dropna()` is applied after concatenating only the variables used in column i , each column's n corresponds to the intersection of nonmissing dates for y and $X_{1:i}$. Later columns can have the same or smaller n depending on whether newly added regressors shorten the intersection. This guarantees that every reported coefficient within a column is estimated on a single, well-defined sample window.

1.3 wedge_var_ratio.py

```
#!/usr/bin/env python3
# -*- coding: utf-8 -*-

import sys
import numpy as np
import pandas as pd
import statsmodels.api as sm

DATE = Date
SPXT = SPXT Index (Natural Log Changes)
BTC = BTC_USD (Natural Log Changes)
CTRL = [
    BCOM Index (Changes) ,
    USTWBGD Index (Changes) ,
    VIX Index (Changes) ,
    USGG10YR Index (Changes) ,
    USGG1M Index (Changes) ,
]

def main(path: str):
    if path.lower().endswith(( '.xlsx' , '.xls ')):
        df = pd.read_excel(path)
    else:
        df = pd.read_csv(path)

    if DATE in df.columns:
        df[DATE] = pd.to_datetime(df[DATE], errors=coerce)
        df = df.sort_values(DATE).reset_index(drop=True)

    for c in [SPXT, BTC] + CTRL:
        df[c] = pd.to_numeric(df[c], errors=coerce)

    y = np.log(df[SPXT]).diff()
    x = np.log(df[BTC]).diff()
    C = pd.DataFrame({c: df[c].diff() for c in CTRL})

    data = pd.concat([y.rename(y), x.rename(x), C], axis=1).dropna()
    if len(data) < 10:
        raise ValueError( Not enough rows after transforms. )

    Xc = sm.add_constant(C.loc[data.index], has_constant= add )
    e_y = sm.OLS(data[ y ], Xc, missing= drop ).fit().resid
    e_x = sm.OLS(data[ x ], Xc, missing= drop ).fit().resid

    wedge = e_y - e_x
    var_ex = float(np.var(e_x, ddof=0))
    if var_ex == 0.0:
        raise ZeroDivisionError( Var(e_x) is zero; cannot compute wedge ratio. )
    wedge_ratio = float(np.var(wedge, ddof=0) / var_ex)
    sqrt_wedge_ratio = float(np.sqrt(wedge_ratio))

    print(f SQRT_WEDGE_VAR_RATIO = {sqrt_wedge_ratio:.6f} )

if __name__ == '__main__':
    if len(sys.argv) < 2:
        print('Usage: python wedge_diagnostics.py program_data.xlsx ')
        sys.exit(1)
    main(sys.argv[1])
```

Objects and transforms. The input file stores *level* series. The script constructs

$$r_t^{s,\text{real}} = \Delta \log(\text{SPXT_Index})_t, \quad r_t^{k,\text{real}} = \Delta \log(\text{BTC_USD})_t,$$

and a control vector

$$C_t = (\Delta \text{BCOM}_t, \Delta \text{USTWBGD}_t, \Delta \text{VIX}_t, \Delta \text{USGG10YR}_t, \Delta \text{USGG1M}_t).$$

After differencing, the code concatenates $(r_t^{s,\text{real}}, r_t^{k,\text{real}}, C_t)$ and calls `dropna()` once, which creates a common sample window.

Orthogonalization to common shocks. With an intercept included, the script fits

$$r_t^{s,\text{real}} = \alpha_s + \beta_s^\top C_t + \varepsilon_t^s, \quad r_t^{k,\text{real}} = \alpha_k + \beta_k^\top C_t + \varepsilon_t^k,$$

and retains only the residuals ε_t^s and ε_t^k . These are the equity and crypto innovations after removing the controls.

Parity wedge and diagnostic. Define the idiosyncratic parity wedge

$$e_t := \varepsilon_t^s - \varepsilon_t^k.$$

The reported statistic is the *square root wedge variance ratio*

$$\sqrt{R} := \sqrt{\frac{\text{Var}(e_t)}{\text{Var}(\varepsilon_t^k)}},$$

computed with the population divisor (`ddof` = 0). If $\text{Var}(\varepsilon_t^k) = 0$, the script raises a division error by design.

Link to the model. Let m_{t+1} denote the real LCU stochastic discount factor. The model implies the risk-adjusted parity

$$\mathbb{E}_t[m_{t+1}((1 + R_{t+1}^{s,\text{real}}) - (1 + R_{t+1}^{k,\text{real}}))] = 0.$$

In short-run regressions, after partialling out common shocks, the Frisch–Waugh–Lovell theorem gives

$$\beta = \frac{\text{Cov}(\varepsilon_t^s, \varepsilon_t^k)}{\text{Var}(\varepsilon_t^k)} = 1 + \frac{\text{Cov}(e_t, \varepsilon_t^k)}{\text{Var}(\varepsilon_t^k)}.$$

By Cauchy–Schwarz,

$$1 - \sqrt{\frac{\text{Var}(e_t)}{\text{Var}(\varepsilon_t^k)}} \leq \beta \leq 1 + \sqrt{\frac{\text{Var}(e_t)}{\text{Var}(\varepsilon_t^k)}}.$$

Hence $\sqrt{R} = \sqrt{\text{Var}(e_t)/\text{Var}(\varepsilon_t^k)}$ directly gauges how tightly β concentrates near 1: small $\sqrt{R}$ implies a strictly positive slope close to unity.

Interpretation. ε_t^k is the crypto leg unexplained by controls. The wedge e_t is the remaining gap between equity and that crypto leg. When $\sqrt{R} \approx 0$, the gap is negligible relative to ε_t^k and the empirical slope β is tightly pinned near 1. Large $\sqrt{R}$ indicates a looser link and a slope that can deviate further from unity.

2 Empirical Analysis

2.1 United States

Table 3: Pre-2020 United States Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{BTC}_{USD})$	0.005** (0.002)	0.003 (0.002)	0.003 (0.002)	-0.002* (0.001)	-0.002* (0.001)	-0.002* (0.001)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.000)	-0.002*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
ΔVIX				-0.005*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10\text{Y Yield}$					0.025*** (0.002)	0.025*** (0.002)
$\Delta 1\text{M Yield}$						-0.001 (0.004)
Observations	3451	3451	3451	3451	3451	3451
Adj. R^2	0.001	0.122	0.146	0.721	0.737	0.737

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 4: Post-2020 United States Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{BTC}_{USD})$	0.135*** (0.007)	0.125*** (0.007)	0.114*** (0.007)	0.036*** (0.005)	0.036*** (0.005)	0.036*** (0.005)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.002*** (0.000)	0.000*** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.007*** (0.001)	-0.004*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10\text{Y Yield}$					0.009*** (0.003)	0.008*** (0.003)
$\Delta 1\text{M Yield}$						0.003 (0.003)
Observations	2087	2087	2087	2087	2087	2087
Adj. R^2	0.145	0.177	0.212	0.646	0.647	0.647

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Interpretation. For the United States, the pre-2020 regressions show that as common-shock controls are added, the Bitcoin slope contracts from a small positive (0.005 in col. 1) to a small negative and weak effect (≈ -0.002 in cols. 4–6), while fit rises sharply (Adj. R^2 : 0.001 $\rightarrow$ 0.737). This pattern is consistent with a “hedge/low comovement” narrative before 2020: once broad macro factors (commodities, dollar, volatility, rates) are accounted for, $\Delta \log(\text{BTC}_{USD})$ contributes little incremental explanatory power for equities. Post-2020, by contrast, the BTC coefficient is large and highly significant in the bivariate specification (0.135, col. 1) and remains economically meaningful after controls (≈ 0.036 in cols. 4–6), with Adj. R^2 improving from 0.145 to ~ 0.647 . This aligns with the paper’s model in which crypto–FX–equity linkages induce stronger equity-like comovement after 2020. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.005$ pre-2020; ≈ 0.990 post-2020), indicating a slightly tighter parity (smaller residual gap) post-2020 and reinforcing the shift from hedge-like behavior to sustained positive co-movement.

Table 5: Pre-2020 United States Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{ETH}_{USD})$	0.005 (0.003)	0.004 (0.003)	0.003 (0.003)	-0.004** (0.002)	-0.004** (0.002)	-0.004** (0.002)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.003*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.003*** (0.001)	-0.001* (0.001)	-0.001** (0.001)	-0.001** (0.001)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10Y \text{ Yield}$					0.025*** (0.004)	0.025*** (0.004)
$\Delta 1M \text{ Yield}$						0.003 (0.005)
Observations	1092	1092	1092	1092	1092	1092
Adj. R^2	0.001	0.064	0.074	0.667	0.679	0.679

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: Post-2020 United States Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{ETH}_{USD})$	0.106*** (0.005)	0.099*** (0.005)	0.090*** (0.005)	0.029*** (0.004)	0.029*** (0.004)	0.029*** (0.004)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.002*** (0.000)	0.000*** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.007*** (0.001)	-0.004*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10\text{Y Yield}$					0.009*** (0.003)	0.009*** (0.003)
$\Delta 1\text{M Yield}$						0.002 (0.003)
Observations	2087	2087	2087	2087	2087	2087
Adj. R^2	0.158	0.189	0.223	0.647	0.648	0.648

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For the United States, the pre-2020 regressions show that as common-shock controls are added, the Ethereum slope contracts from a small, imprecise positive (0.005 in col. 1) to a statistically significant negative and weak effect (≈ -0.004 in cols. 4–6), while fit rises sharply (Adj. R^2 : 0.001 $\rightarrow$ 0.679). This pattern is consistent with a “hedge/low comovement” narrative before 2020: once broad macro factors (commodities, dollar, volatility, rates) are accounted for, $\Delta \log(\text{ETH}_{USD})$ contributes little incremental explanatory power for equities. Post-2020, by contrast, the ETH coefficient is large and highly significant in the bivariate specification (0.106, col. 1) and remains economically meaningful after controls (≈ 0.029 in cols. 4–6), with Adj. R^2 improving from 0.158 to ~ 0.648 . This aligns with the paper’s model in which crypto–FX–equity linkages induce stronger equity-like comovement after 2020. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.006$ pre-2020; ≈ 0.986 post-2020), indicating a slightly tighter parity (smaller residual gap) post-2020 and reinforcing the shift from hedge-like behavior to sustained positive co-movement.

Table 7: Pre-2020 United States Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{LTC}_{USD})$	0.005 (0.006)	0.005 (0.005)	0.004 (0.005)	-0.000 (0.003)	-0.000 (0.003)	-0.001 (0.003)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
ΔDollar			-0.006*** (0.001)	-0.002** (0.001)	-0.001** (0.001)	-0.001** (0.001)
ΔVIX				-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
$\Delta 10\text{Y Yield}$					0.024*** (0.005)	0.024*** (0.005)
$\Delta 1\text{M Yield}$						0.003 (0.006)
Observations	691	691	691	691	691	691
Adj. R^2	-0.000	0.089	0.111	0.742	0.750	0.750

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 8: Post-2020 United States Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{LTC}_{USD})$	0.074*** (0.005)	0.068*** (0.005)	0.062*** (0.005)	0.015*** (0.003)	0.015*** (0.003)	0.015*** (0.003)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.007*** (0.001)	-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10\text{Y Yield}$					0.009*** (0.003)	0.009*** (0.003)
$\Delta 1\text{M Yield}$						0.002 (0.003)
Observations	2087	2087	2087	2087	2087	2087
Adj. R^2	0.096	0.136	0.180	0.640	0.642	0.642

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Interpretation. For the United States, the pre-2020 regressions show that as common-shock controls are added, the Litecoin slope contracts from a small positive (0.005 in col. 1) to effectively zero (≈ 0 in cols. 4–6), while fit rises sharply (Adj. R^2 : 0.000 $\rightarrow$ 0.750). This pattern is consistent with a “hedge/low comovement” narrative before 2020: once broad macro factors (commodities, dollar, volatility, rates) are accounted for, $\Delta \log(\text{LTC}_{USD})$ contributes little incremental explanatory power for equities. Post-2020, by contrast, the LTC coefficient is large and highly significant in the bivariate specification (0.074, col. 1) and remains economically meaningful after controls (≈ 0.015 in cols. 4–6), with Adj. R^2 improving from 0.096 to ~ 0.642 . This aligns with the model’s crypto–FX–equity linkage: as common shocks are netted out, crypto loads more like an equity factor in the later period. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.003$ pre-2020; ≈ 0.997 post-2020), indicating a slightly tighter parity (smaller residual gap) post-2020 and reinforcing the shift from hedge-like behavior to sustained positive co-movement.

Table 9: Pre-2020 United States Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{USDT}_{USD})$	0.009 (0.028)	0.018 (0.027)	0.016 (0.027)	0.003 (0.016)	0.005 (0.016)	0.004 (0.016)
$\Delta \text{Commodity}$		0.004*** (0.001)	0.004*** (0.001)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.004*** (0.001)	-0.002** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10\text{Y Yield}$					0.026*** (0.005)	0.026*** (0.005)
$\Delta 1\text{M Yield}$						0.003 (0.006)
Observations	879	879	879	879	879	879
Adj. R^2	-0.001	0.076	0.092	0.677	0.688	0.688

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For the United States, the pre-2020 regressions show that as common-shock controls are added, the Tether slope stays small and statistically weak (from 0.009 in col. 1 to ≈ 0.003 – 0.005 in cols. 4–6), while fit rises sharply (Adj. R^2 : $-0.001 \rightarrow 0.688$). This is exactly what a USD–pegged asset implies: once broad USD–linked drivers (commodities, dollar, volatility, rates) are partialled out, there is essentially no independent variation left for $\Delta \log(\text{USDT}_{USD})$ to explain; any residual slope likely reflects peg frictions or microstructure noise rather than an equity risk loading. Post-2020, the raw bivariate coefficient is imprecisely negative (-0.216 , col. 1) but, after controls, again shrinks toward zero (≈ -0.03 in cols. 4–6) with fit improving to ~ 0.638 . Using the square-root variance ratio $\sqrt{\rho}$, the windows differ markedly ($\sqrt{\rho} \approx 1.100$ pre-2020; ≈ 6.586 post-2020); the large post-2020 value is consistent with residual dispersion from peg mechanics rather than genuine equity co-movement, reinforcing the near-zero slope narrative.

Interpretation. For the United States, the pre-2020 regressions show that as common-shock controls are added, the Ripple slope shrinks from a small, imprecise positive (0.006 in col. 1) to effectively zero (≈ 0 in cols. 4–6), while fit rises sharply (Adj. R^2 : 0.000 $\rightarrow$ 0.750). This pattern is consistent with a “hedge/low comovement” narrative before 2020: once broad macro factors (commodities, dollar, volatility, rates) are accounted for, $\Delta \log(\text{XRP}_{USD})$ contributes little incremental explanatory power for equities. Post-2020, by contrast, the XRP coefficient is positive and highly significant in the bivariate specification (0.049, col. 1) and remains economically meaningful after controls (≈ 0.013 in cols. 4–6), with Adj. R^2 improving from

Table 10: Post-2020 United States Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{USDT}_{USD})$	-0.216 (0.237)	-0.159 (0.230)	-0.114 (0.223)	-0.022 (0.143)	-0.031 (0.143)	-0.030 (0.143)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10Y \text{ Yield}$					0.009*** (0.003)	0.009*** (0.003)
$\Delta 1M \text{ Yield}$						0.002 (0.003)
Observations	2087	2087	2087	2087	2087	2087
Adj. R^2	-0.000	0.056	0.115	0.637	0.638	0.638

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 11: Pre-2020 United States Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{XRP}_{USD})$	0.006 (0.006)	0.006 (0.006)	0.006 (0.006)	-0.000 (0.003)	-0.000 (0.003)	-0.000 (0.003)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
ΔDollar			-0.006*** (0.001)	-0.002** (0.001)	-0.001** (0.001)	-0.001** (0.001)
ΔVIX				-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
$\Delta 10Y \text{ Yield}$					0.024*** (0.005)	0.024*** (0.005)
$\Delta 1M \text{ Yield}$						0.003 (0.006)
Observations	691	691	691	691	691	691
Adj. R^2	0.000	0.089	0.111	0.742	0.750	0.750

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 12: Post-2020 United States Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{XRP}_{USD})$	0.049*** (0.005)	0.044*** (0.004)	0.041*** (0.004)	0.013*** (0.003)	0.013*** (0.003)	0.013*** (0.003)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.008*** (0.001)	-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)
ΔVIX				-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
$\Delta 10\text{Y Yield}$					0.009*** (0.003)	0.009*** (0.003)
$\Delta 1\text{M Yield}$						0.002 (0.003)
Observations	2087	2087	2087	2087	2087	2087
Adj. R^2	0.053	0.099	0.152	0.640	0.642	0.642

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

0.053 to ~ 0.642 . This aligns with the paper's model in which crypto-FX-equity linkages induce stronger equity-like comovement after 2020. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.003$ pre-2020; ≈ 0.996 post-2020), indicating a slightly tighter parity (smaller residual gap) post-2020 and reinforcing the shift from hedge-like behavior to sustained positive co-movement.

2.2 United Kingdom

Table 13: Pre-2020 United Kingdom Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{GBP})$	0.003 (0.002)	0.002 (0.002)	0.003 (0.002)	0.000 (0.002)	0.000 (0.002)	0.000 (0.002)	0.000 (0.002)	0.000 (0.002)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.005*** (0.000)	-0.003*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)	-0.004*** (0.000)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.025*** (0.003)	0.025*** (0.003)	0.022*** (0.003)	0.022*** (0.003)
$\Delta 1\text{M Yield}$						0.010* (0.006)	0.010* (0.006)	0.010* (0.006)
$\Delta 10\text{Y Spread}$							0.007** (0.003)	0.007** (0.003)
$\Delta 3\text{M Spread}$								-0.009* (0.005)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	0.000	0.137	0.160	0.307	0.322	0.322	0.323	0.323

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 14: Post-2020 United Kingdom Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{GBP})$	0.059*** (0.006)	0.051*** (0.006)	0.045*** (0.006)	0.018*** (0.006)	0.016*** (0.006)	0.016*** (0.006)	0.016*** (0.006)	0.016*** (0.006)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.008*** (0.001)	-0.007*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.030*** (0.003)	0.030*** (0.003)	0.036*** (0.004)	0.036*** (0.004)
$\Delta 1\text{M Yield}$						0.008** (0.003)	0.008** (0.003)	0.008** (0.003)
$\Delta 10\text{Y Spread}$							-0.012*** (0.003)	-0.012*** (0.003)
$\Delta 3\text{M Spread}$								-0.001 (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.042	0.098	0.189	0.272	0.302	0.303	0.308	0.308

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For the United Kingdom, the pre-2020 regressions show that as common-shock controls are added, the Bitcoin slope contracts from a small, imprecise positive (0.003 in col. 1) to effectively zero (≈ 0 in cols. 4–8), while fit rises (Adj. R^2 : 0.000 $\rightarrow$ 0.323). This pattern is consistent with a “hedge/low co-movement” narrative before 2020: once broad macro factors (commodities, dollar, volatility, rates, and term spreads) are accounted for, $\Delta \log(\text{BTC}_{GBP})$ contributes little incremental explanatory power for equities. Post-2020, by contrast, the BTC coefficient is positive and highly significant in the bivariate specification (0.059, col. 1) and remains economically meaningful after controls (≈ 0.016 in cols. 5–8), with Adj. R^2 improving from 0.042 to ~ 0.308 . This aligns with the paper’s model in which crypto–FX–equity linkages induce stronger equity-like comovement after 2020. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.006$ pre-2020; ≈ 1.017 post-2020), indicating a modestly looser parity (slightly larger residual gap) post-2020, while the slope dynamics still point to a shift from hedge-like behavior to sustained positive co-movement.

Interpretation. For the United Kingdom, the pre-2020 regressions show that as common-shock controls are added, the Ethereum slope remains small but persistently positive (from 0.009 in col. 1 to ≈ 0.005 in cols. 7–8), while fit increases steadily (Adj. R^2 : 0.007 $\rightarrow$ 0.187). This contrasts with the pre-2020 BTC pattern by preserving a modest equity-like signal even after accounting for broad macro factors (commodities, dollar, volatility, rates, and term spreads), though the effect is economically small relative to the contribution of the controls. Post-2020, by contrast, the ETH coefficient is large and highly significant in the bivariate specification (0.048, col. 1) and remains economically meaningful after controls (≈ 0.012 – 0.013 in cols. 4–8), with Adj. R^2 improving from 0.048 to ~ 0.308 . This aligns with the paper’s model in which crypto–FX–equity linkages induce stronger equity-like co-movement after 2020. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 0.999$ pre-2020; ≈ 1.006 post-2020), indicating similar residual parity across periods; together with the rising slopes, this supports a transition from weak to sustained positive co-movement.

Interpretation. For the United Kingdom, the fully controlled Litecoin slopes are essentially the same across periods. Pre-2020, the LTC coefficient settles around ≈ 0.006 once common-shock controls are included (cols. 6–8) and is not statistically significant, while fit rises from Adj. $R^2 = 0.003$ to 0.236. Post-2020, the bivariate loading is larger (0.031, col. 1), but after adding the same controls the slope again lands near ≈ 0.006 (cols. 7–8), now marginally significant. Adj. R^2 improves from 0.025 to ~ 0.306 as controls are

Table 15: Pre-2020 United Kingdom Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{GBP})$	0.009*** (0.003)	0.008*** (0.003)	0.008*** (0.003)	0.006** (0.003)	0.006** (0.003)	0.006** (0.003)	0.005* (0.003)	0.005* (0.003)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.004*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
ΔDollar			0.001 (0.001)	0.001* (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.015*** (0.005)	0.014*** (0.005)	0.024*** (0.006)	0.024*** (0.006)
$\Delta 1\text{M Yield}$						0.015** (0.007)	0.016** (0.007)	0.016** (0.007)
$\Delta 10\text{Y Spread}$							-0.019*** (0.005)	-0.019*** (0.005)
$\Delta 3\text{M Spread}$								-0.010* (0.006)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	0.007	0.082	0.082	0.168	0.173	0.176	0.186	0.187

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 16: Post-2020 United Kingdom Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{GBP})$	0.048*** (0.005)	0.041*** (0.005)	0.035*** (0.004)	0.013*** (0.004)	0.012*** (0.004)	0.012*** (0.004)	0.013*** (0.004)	0.013*** (0.004)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.008*** (0.001)	-0.007*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.031*** (0.003)	0.030*** (0.003)	0.037*** (0.004)	0.037*** (0.004)
$\Delta 1\text{M Yield}$						0.008** (0.003)	0.008** (0.003)	0.008** (0.003)
$\Delta 10\text{Y Spread}$							-0.012*** (0.003)	-0.012*** (0.003)
$\Delta 3\text{M Spread}$								-0.001 (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.048	0.103	0.190	0.272	0.302	0.304	0.308	0.308

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 17: Pre-2020 United Kingdom Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{GBP})$	0.008* (0.005)	0.008* (0.004)	0.009* (0.004)	0.007 (0.004)	0.007 (0.004)	0.006 (0.004)	0.006 (0.004)	0.006 (0.004)
$\Delta \text{Commodity}$		0.005*** (0.000)	0.004*** (0.001)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
ΔDollar			-0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.016** (0.007)	0.014* (0.007)	0.022*** (0.008)	0.022*** (0.008)
$\Delta 1\text{M Yield}$						0.023** (0.009)	0.023*** (0.009)	0.023*** (0.009)
$\Delta 10\text{Y Spread}$							-0.017** (0.007)	-0.017** (0.007)
$\Delta 3\text{M Spread}$								0.005 (0.010)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.003	0.122	0.121	0.220	0.224	0.230	0.236	0.236

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 18: Post-2020 United Kingdom Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{GBP})$	0.031*** (0.004)	0.026*** (0.004)	0.023*** (0.004)	0.006 (0.004)	0.006 (0.004)	0.006 (0.004)	0.006* (0.004)	0.006* (0.004)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.031*** (0.003)	0.030*** (0.003)	0.037*** (0.004)	0.037*** (0.004)
$\Delta 1\text{M Yield}$						0.008** (0.003)	0.008** (0.003)	0.008** (0.003)
$\Delta 10\text{Y Spread}$							-0.012*** (0.003)	-0.012*** (0.003)
$\Delta 3\text{M Spread}$								-0.001 (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.025	0.086	0.179	0.270	0.300	0.302	0.306	0.306

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

added, underscoring that broad macro factors (commodities, dollar, volatility, rates, term spreads) account for most equity variation in both windows. Using the square-root variance ratio $\sqrt{\rho}$, both periods lie near unity ($\sqrt{\rho} \approx 1.000$ pre-2020; ≈ 1.008 post-2020), so the narrative shift here is about robustness and significance—plus a larger bivariate loading—rather than a higher fully controlled beta.

Table 19: Pre-2020 United Kingdom Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{GBP})$	0.037* (0.021)	0.057*** (0.020)	0.059*** (0.021)	0.056*** (0.020)	0.057*** (0.020)	0.056*** (0.020)	0.046** (0.020)	0.046** (0.020)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.004*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
ΔDollar			-0.000 (0.001)	0.001 (0.001)	0.001 (0.001)	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.013** (0.006)	0.012* (0.006)	0.018*** (0.007)	0.019*** (0.007)
$\Delta 1\text{M Yield}$						0.014* (0.008)	0.014* (0.008)	0.014* (0.008)
$\Delta 10\text{Y Spread}$							-0.012** (0.006)	-0.012** (0.006)
$\Delta 3\text{M Spread}$								-0.011 (0.007)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	0.002	0.097	0.096	0.183	0.186	0.188	0.191	0.192

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 20: Post-2020 United Kingdom Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{GBP})$	-0.302*** (0.039)	-0.184*** (0.040)	0.312*** (0.048)	0.346*** (0.045)	0.282*** (0.045)	0.280*** (0.045)	0.272*** (0.045)	0.272*** (0.045)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.012*** (0.001)	-0.011*** (0.001)	-0.011*** (0.001)	-0.011*** (0.001)	-0.011*** (0.001)	-0.011*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.027*** (0.003)	0.027*** (0.003)	0.033*** (0.004)	0.033*** (0.004)
$\Delta 1\text{M Yield}$						0.007** (0.003)	0.007** (0.003)	0.007** (0.003)
$\Delta 10\text{Y Spread}$							-0.011*** (0.003)	-0.011*** (0.003)
$\Delta 3\text{M Spread}$								-0.001 (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.027	0.077	0.182	0.289	0.312	0.314	0.317	0.317

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For the United Kingdom, the pre-2020 regressions show that as common-shock controls are added, the Tether slope settles at a small but statistically significant positive level (from 0.037 in col. 1 to ≈ 0.046 in cols. 7–8), while fit rises (Adj. R^2 : 0.002 $\rightarrow$ 0.192). Post-2020, the bivariate USDT coefficient

is negative (-0.302 , col. 1) but flips and becomes strongly positive once core controls enter (about 0.27 in cols. 7–8), with Adj. R^2 improving from 0.027 to ~ 0.317 . This pattern is consistent with the paper’s model in a GBP setting: because USDT is effectively a USD proxy, omitting dollar/volatility/rate controls induces sign distortions in the raw regression, whereas after netting out common shocks the stablecoin inherits an equity-like loading via the FX channel in the post-2020 period. Using the square-root variance ratio $\sqrt{\rho}$, dispersion is modest pre-2020 and elevated post-2020 ($\sqrt{\rho} \approx 1.126$ pre; ≈ 2.171 post), consistent with peg mechanics in GBP terms; nevertheless, the fully controlled post-2020 slope remains high, reinforcing the shift toward positive co-movement once macro factors are accounted for.

Table 21: Pre-2020 United Kingdom Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{GBP})$	0.010** (0.005)	0.011** (0.005)	0.011** (0.005)	0.009** (0.005)	0.009* (0.004)	0.009* (0.004)	0.008* (0.004)	0.008* (0.004)
$\Delta \text{Commodity}$		0.005*** (0.000)	0.004*** (0.001)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
ΔDollar			-0.001 (0.001)	0.000 (0.001)	0.001 (0.001)	0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.016** (0.007)	0.014* (0.007)	0.022*** (0.008)	0.022*** (0.008)
$\Delta 1\text{M Yield}$						0.023** (0.009)	0.023*** (0.009)	0.023*** (0.009)
$\Delta 10\text{Y Spread}$							-0.016** (0.007)	-0.016** (0.007)
$\Delta 3\text{M Spread}$								0.006 (0.010)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.004	0.124	0.123	0.221	0.226	0.232	0.237	0.237

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 22: Post-2020 United Kingdom Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{GBP})$	0.017*** (0.004)	0.014*** (0.004)	0.013*** (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.031*** (0.003)	0.030*** (0.003)	0.037*** (0.004)	0.037*** (0.004)
$\Delta 1\text{M Yield}$						0.008** (0.003)	0.008** (0.003)	0.008** (0.003)
$\Delta 10\text{Y Spread}$							-0.012*** (0.003)	-0.012*** (0.003)
$\Delta 3\text{M Spread}$								-0.001 (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.009	0.073	0.171	0.269	0.299	0.301	0.306	0.305

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For the United Kingdom, the pre-2020 regressions show that as common-shock controls are added, the Ripple slope remains small but persistently positive (from 0.010 in col. 1 to ≈ 0.008 in cols. 7–8), while fit rises (Adj. R^2 : 0.004 $\rightarrow$ 0.237). This is consistent with a modest equity-like signal that survives after accounting for broad macro factors (commodities, dollar, volatility, rates, and term spreads), albeit at a small magnitude. Post-2020, by contrast, the bivariate XRP coefficient is larger (0.017, col. 1) but collapses to a statistically insignificant ≈ 0.003 once the same controls are included (cols. 4–8), with Adj. R^2 improving from 0.009 to ~ 0.305 . Taken together, the tables indicate that any apparent comovement in the later period is largely absorbed by common shocks, leaving little incremental role for $\Delta \log(\text{XRP}_{GBP})$ once controls are present. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 0.999$ pre-2020; ≈ 1.008 post-2020), indicating similar residual parity across periods and reinforcing that XRP’s equity-like loading in the U.K. is small and not robust after 2020.

2.3 Eurozone

Table 23: Pre-2020 Eurozone Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{EUR})$	0.003 (0.003)	0.002 (0.002)	0.003 (0.002)	-0.000 (0.002)	-0.000 (0.002)	-0.000 (0.002)	-0.001 (0.002)	-0.001 (0.002)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
ΔVIX				-0.003*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.032*** (0.003)	0.031*** (0.003)	0.008** (0.004)	0.007** (0.004)
$\Delta 1\text{M Yield}$						0.015** (0.006)	0.013** (0.006)	0.013** (0.006)
$\Delta 10\text{Y Spread}$							0.048*** (0.004)	0.048*** (0.004)
$\Delta 3\text{M Spread}$								-0.015** (0.006)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	0.000	0.111	0.129	0.315	0.336	0.337	0.363	0.364

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For the Eurozone, the pre-2020 regressions show that as common-shock controls are added, the Bitcoin slope fades from a small positive (0.003 in col. 1) to essentially zero (about 0 to -0.001 in cols. 4–8), while fit rises (Adj. R^2 : 0.000 $\rightarrow$ 0.364). This supports a “hedge/low comovement” narrative before 2020: once broad macro factors (commodities, dollar, volatility, rates, and term spreads) are accounted for, $\Delta \log(\text{BTC}_{EUR})$ offers little incremental explanatory power for Eurozone equities. Post-2020, by contrast, the BTC coefficient is large and highly significant in the bivariate specification (0.072, col. 1) and remains economically meaningful after controls (≈ 0.022 in cols. 4–8), with Adj. R^2 improving from 0.061 to ~ 0.359 . This aligns with the paper’s model in which crypto–FX–equity linkages generate stronger equity-like co-movement after 2020. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.007$ pre-2020; ≈ 1.009 post-2020), indicating similar residual parity across periods and reinforcing that the post-2020 change is driven by higher BTC loadings rather than shifts in residual dispersion.

Interpretation. For the Eurozone, the pre-2020 regressions show that as common-shock controls are added, the Ethereum slope shrinks from a small positive (0.005 in col. 1) to an economically negligible value (≈ 0.001 in cols. 4–8), while fit rises (Adj. R^2 : 0.002 $\rightarrow$ 0.232). This supports a pre-2020 “hedge/low

Table 24: Post-2020 Eurozone Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{EUR})$	0.072*** (0.006)	0.064*** (0.006)	0.055*** (0.006)	0.023*** (0.006)	0.022*** (0.005)	0.022*** (0.005)	0.022*** (0.005)	0.022*** (0.005)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.010*** (0.001)	-0.009*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.022*** (0.003)	0.021*** (0.003)	0.020*** (0.004)	0.021*** (0.004)
$\Delta 1\text{M Yield}$						0.008*** (0.003)	0.008*** (0.003)	0.008*** (0.003)
$\Delta 10\text{Y Spread}$							0.002 (0.005)	0.000 (0.005)
$\Delta 3\text{M Spread}$								0.013* (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.061	0.107	0.239	0.342	0.356	0.358	0.358	0.359

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 25: Pre-2020 Eurozone Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{EUR})$	0.005* (0.003)	0.005 (0.003)	0.005 (0.003)	0.001 (0.003)	0.001 (0.003)	0.001 (0.003)	0.001 (0.003)	0.001 (0.003)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.000 (0.001)	0.001 (0.001)	0.000 (0.001)	0.000 (0.001)	0.001 (0.001)	0.001 (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.020*** (0.005)	0.020*** (0.005)	-0.001 (0.007)	-0.001 (0.007)
$\Delta 1\text{M Yield}$						0.016** (0.006)	0.012** (0.006)	0.013** (0.006)
$\Delta 10\text{Y Spread}$							0.040*** (0.008)	0.040*** (0.008)
$\Delta 3\text{M Spread}$								0.008 (0.010)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	0.002	0.056	0.055	0.201	0.212	0.215	0.232	0.232

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 26: Post-2020 Eurozone Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{EUR})$	0.061*** (0.005)	0.054*** (0.005)	0.046*** (0.004)	0.021*** (0.004)	0.021*** (0.004)	0.021*** (0.004)	0.021*** (0.004)	0.020*** (0.004)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.010*** (0.001)	-0.009*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.022*** (0.003)	0.021*** (0.003)	0.021*** (0.004)	0.021*** (0.004)
$\Delta 1\text{M Yield}$						0.008*** (0.003)	0.008*** (0.003)	0.008*** (0.003)
$\Delta 10\text{Y Spread}$							0.001 (0.005)	-0.001 (0.005)
$\Delta 3\text{M Spread}$								0.012* (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.076	0.120	0.246	0.344	0.359	0.361	0.360	0.361

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

comovement” narrative in which broad macro factors (commodities, dollar, volatility, rates, term spreads) absorb most equity variation, leaving $\Delta \log(\text{ETH}_{EUR})$ with little incremental power. Post-2020, by contrast, the ETH coefficient is large and highly significant in the bivariate specification (0.061, col. 1) and remains economically meaningful after controls (≈ 0.020 – 0.021 in cols. 4–8), with Adj. R^2 improving from 0.076 to ~ 0.361 . This aligns with the paper’s model in which crypto–FX–equity linkages generate stronger equity-like co-movement after 2020. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.003$ pre-2020; ≈ 0.998 post-2020), indicating similar residual parity across periods and reinforcing that the post-2020 shift is driven by higher ETH loadings rather than changes in residual dispersion.

Table 27: Pre-2020 Eurozone Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{EUR})$	0.012** (0.005)	0.012*** (0.004)	0.012*** (0.004)	0.010** (0.004)	0.010** (0.004)	0.009** (0.004)	0.010** (0.004)	0.010** (0.004)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.004*** (0.001)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.003*** (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.016** (0.007)	0.014** (0.007)	-0.007 (0.009)	-0.007 (0.009)
$\Delta 1\text{M Yield}$						0.025*** (0.008)	0.022** (0.008)	0.022*** (0.008)
$\Delta 10\text{Y Spread}$							0.042*** (0.011)	0.042*** (0.011)
$\Delta 3\text{M Spread}$								0.017 (0.014)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.008	0.109	0.117	0.271	0.276	0.284	0.298	0.298

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 28: Post-2020 Eurozone Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{EUR})$	0.036*** (0.004)	0.031*** (0.004)	0.027*** (0.004)	0.007* (0.004)	0.007* (0.004)	0.007* (0.004)	0.007* (0.004)	0.007* (0.004)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001** (0.000)	0.001** (0.000)	0.001*** (0.000)
ΔDollar			-0.011*** (0.001)	-0.009*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.022*** (0.003)	0.021*** (0.003)	0.021*** (0.004)	0.022*** (0.004)
$\Delta 1\text{M Yield}$						0.008*** (0.003)	0.008*** (0.003)	0.008*** (0.003)
$\Delta 10\text{Y Spread}$							0.000 (0.005)	-0.001 (0.005)
$\Delta 3\text{M Spread}$								0.013* (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.034	0.086	0.222	0.337	0.352	0.354	0.354	0.355

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For the Eurozone, the pre-2020 regressions show that as common-shock controls are added, the Litecoin slope remains small yet persistently positive (from 0.012 in col. 1 to ≈ 0.010 in cols. 4–8), while fit rises (Adj. R^2 : 0.008 $\rightarrow$ 0.298). This indicates a modest equity-like signal that survives after accounting for broad macro factors (commodities, dollar, volatility, rates, term spreads), albeit at a limited magnitude. Post-2020, the bivariate coefficient is larger (0.036, col. 1) but compresses to a small, still significant ≈ 0.007 once controls are included (cols. 4–8), with Adj. R^2 improving from 0.034 to ~ 0.355 . Taken together, the tables suggest that most comovement is absorbed by common shocks in both windows, while $\Delta \log(\text{LTC}_{EUR})$ retains a modest residual loading that remains positive after 2020. Using the square-root variance ratio $\sqrt{\rho}$, both periods lie near unity ($\sqrt{\rho} \approx 0.996$ pre-2020; ≈ 1.007 post-2020), pointing to similar residual parity and reinforcing that the principal change over time is in overall fit and significance rather than a materially larger fully controlled beta.

Interpretation. For the Eurozone, the pre-2020 regressions show that as common-shock controls are added, the Tether slope remains small and statistically weak (from 0.002 in col. 1 to ≈ 0.023 – 0.025 in cols. 7–8, without significance), while fit rises (Adj. R^2 : $-0.001 \rightarrow 0.239$). This is consistent with a near-neutral/low-comovement narrative once broad macro factors (commodities, dollar, volatility, rates, term spreads) are accounted for. Post-2020, however, the bivariate coefficient is strongly negative (-0.319 , col. 1) but flips to a large, highly significant positive loading after controls (≈ 0.45 in cols. 7–8), with Adj. R^2 improving from 0.021 to ~ 0.374 . In the paper’s framework, this pattern reflects USDT’s role as a USD proxy in EUR space: omitting dollar/rate/volatility variables induces sign distortions, whereas once common shocks are netted out, the residual exposure co-moves positively with equities via the FX/crypto channel in the later period. Using the square-root variance ratio $\sqrt{\rho}$, dispersion is modest pre-2020 and elevated post-2020 ($\sqrt{\rho} \approx 1.137$ pre; ≈ 2.528 post), consistent with peg mechanics and EUR/USD microstructure; nevertheless, the fully controlled post-2020 slope remains high, indicating substantive positive co-movement after controls.

Interpretation. For the Eurozone, the pre-2020 regressions show that as common-shock controls are added, the Ripple slope remains small but positive (from 0.009 in col. 1 to ≈ 0.007 – 0.008 in cols. 4–8, marginally significant), while fit rises (Adj. R^2 : 0.004 $\rightarrow$ 0.296). This indicates a modest equity-like signal that survives after accounting for broad macro factors (commodities, dollar, volatility, rates, term spreads), albeit at limited magnitude. Post-2020, the bivariate XRP coefficient is larger (0.020, col. 1) but compresses

Table 29: Pre-2020 Eurozone Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{EUR})$	0.002 (0.022)	0.019 (0.021)	0.026 (0.022)	0.016 (0.020)	0.016 (0.020)	0.015 (0.020)	0.025 (0.020)	0.023 (0.020)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.002* (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.001 (0.001)	0.000 (0.001)	0.000 (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.018*** (0.006)	0.017*** (0.006)	-0.006 (0.008)	-0.006 (0.008)
$\Delta 1\text{M Yield}$						0.017** (0.007)	0.014* (0.007)	0.014* (0.007)
$\Delta 10\text{Y Spread}$							0.045*** (0.010)	0.045*** (0.010)
$\Delta 3\text{M Spread}$								0.012 (0.012)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	-0.001	0.072	0.074	0.210	0.217	0.221	0.239	0.239

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 30: Post-2020 Eurozone Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{EUR})$	-0.319*** (0.047)	-0.238*** (0.047)	0.605*** (0.057)	0.484*** (0.053)	0.438*** (0.053)	0.432*** (0.053)	0.451*** (0.054)	0.451*** (0.054)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)
ΔDollar			-0.016*** (0.001)	-0.014*** (0.001)	-0.014*** (0.001)	-0.014*** (0.001)	-0.014*** (0.001)	-0.014*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.018*** (0.003)	0.018*** (0.003)	0.013*** (0.004)	0.014*** (0.004)
$\Delta 1\text{M Yield}$						0.007** (0.003)	0.007** (0.003)	0.007** (0.003)
$\Delta 10\text{Y Spread}$							0.008* (0.005)	0.006 (0.005)
$\Delta 3\text{M Spread}$								0.014** (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.021	0.072	0.244	0.362	0.372	0.373	0.373	0.374

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 31: Pre-2020 Eurozone Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{EUR})$	0.009* (0.005)	0.010** (0.005)	0.010** (0.005)	0.007* (0.004)	0.007* (0.004)	0.007 (0.004)	0.008* (0.004)	0.007* (0.004)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.004*** (0.001)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.003*** (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.016** (0.007)	0.014** (0.007)	-0.007 (0.009)	-0.007 (0.009)
$\Delta 1\text{M Yield}$						0.025*** (0.008)	0.022*** (0.008)	0.023*** (0.008)
$\Delta 10\text{Y Spread}$							0.042*** (0.011)	0.042*** (0.011)
$\Delta 3\text{M Spread}$								0.017 (0.014)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.004	0.105	0.113	0.268	0.273	0.281	0.295	0.296

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 32: Post-2020 Eurozone Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{EUR})$	0.020*** (0.004)	0.017*** (0.004)	0.016*** (0.003)	0.004 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.011*** (0.001)	-0.009*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.022*** (0.003)	0.021*** (0.003)	0.021*** (0.004)	0.022*** (0.004)
$\Delta 1\text{M Yield}$						0.008*** (0.003)	0.008*** (0.003)	0.008** (0.003)
$\Delta 10\text{Y Spread}$							0.000 (0.005)	-0.001 (0.005)
$\Delta 3\text{M Spread}$								0.013* (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.013	0.070	0.212	0.337	0.351	0.353	0.353	0.354

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

to an economically small and statistically weak ≈ 0.003 once controls are included (cols. 4–8), with Adj. R^2 improving from 0.013 to ~ 0.354 . Taken together, the tables suggest that most of XRP’s apparent comovement in the later period is absorbed by common shocks, leaving little incremental role for $\Delta \log(\text{XRP}_{EUR})$ once controls are present. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 0.9987$ pre-2020; ≈ 1.0070 post-2020), pointing to similar residual parity and reinforcing that the principal change is improved overall fit rather than a materially larger fully controlled beta.

2.4 Japan

Table 33: Pre-2020 Japan Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{JPY})$	0.004 (0.003)	0.003 (0.003)	0.003 (0.003)	0.002 (0.003)	0.001 (0.003)	0.001 (0.003)	-0.001 (0.003)	-0.001 (0.003)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.001 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.012** (0.005)	0.012** (0.005)	0.006 (0.005)	0.006 (0.005)
$\Delta 1\text{M Yield}$						-0.011 (0.010)	-0.012 (0.009)	-0.012 (0.009)
$\Delta 10\text{Y Spread}$							0.163*** (0.012)	0.163*** (0.012)
$\Delta 3\text{M Spread}$								0.019** (0.009)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	0.000	0.011	0.011	0.019	0.020	0.020	0.072	0.073

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For Japan, the pre-2020 regressions show the Bitcoin slope drifting from a small positive (0.004 in col. 1) to essentially zero/slightly negative by the fully controlled specs (≈ -0.001 in cols. 7–8), while fit rises only modestly (Adj. R^2 : 0.000 $\rightarrow$ 0.073). This points to a weak hedge/low-comovement pattern before 2020 once broad macro factors are netted out; notably, the term-structure proxies (10Y spread ≈ 0.163 ***; 3M spread 0.019**) explain far more of the variation than $\Delta \log(\text{BTC}_{JPY})$ itself. Post-2020, the bivariate BTC coefficient is larger (0.052, col. 1) and remains economically meaningful after controls (≈ 0.031 – 0.034 in cols. 4–8), but overall fit still improves only slightly (Adj. R^2 : 0.020 $\rightarrow$ 0.094), suggesting that common shocks capture most equity dynamics and that Japan’s equity–crypto linkage strengthens yet remains modest relative to the U.S./Eurozone. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.018$ pre-2020; ≈ 1.034 post-2020), consistent with similar residual parity and a narrative in which the post-2020 shift reflects somewhat higher BTC loadings rather than changes in residual dispersion.

Interpretation. For Japan, the pre-2020 regressions show that as common-shock controls are added, the Ethereum slope moves from a tiny positive (0.002 in col. 1) to essentially zero (≈ 0 in cols. 4–8), while fit rises modestly (Adj. R^2 : $-0.001 \rightarrow 0.117$). This is consistent with a low-comovement narrative: once broad macro factors (commodities, dollar, volatility, rates) are accounted for, $\Delta \log(\text{ETH}_{JPY})$ contributes little incremental explanatory power for equities. Post-2020, by contrast, the bivariate ETH coefficient is materially larger and highly significant (0.045, col. 1) and remains economically meaningful after controls (≈ 0.030 in cols. 4–8), with Adj. R^2 improving from 0.027 to ~ 0.096 . Using the square-root variance ratio

Table 34: Post-2020 Japan Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{JPY})$	0.052*** (0.008)	0.045*** (0.008)	0.041*** (0.008)	0.033*** (0.008)	0.031*** (0.008)	0.031*** (0.008)	0.034*** (0.008)	0.034*** (0.008)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.004*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.009* (0.005)	0.009* (0.005)	0.004 (0.005)	0.004 (0.005)
$\Delta 1M \text{ Yield}$						-0.000 (0.005)	0.001 (0.005)	0.001 (0.005)
$\Delta 10Y \text{ Spread}$							0.129*** (0.013)	0.128*** (0.013)
$\Delta 3M \text{ Spread}$								0.009 (0.014)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.020	0.033	0.050	0.053	0.055	0.054	0.094	0.094

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 35: Pre-2020 Japan Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{JPY})$	0.002 (0.004)	0.001 (0.004)	0.001 (0.004)	0.000 (0.004)	-0.000 (0.004)	-0.000 (0.004)	0.000 (0.004)	0.001 (0.004)
$\Delta \text{Commodity}$		0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.001 (0.001)	-0.001 (0.001)
ΔVIX				-0.000** (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000** (0.000)	-0.000** (0.000)
$\Delta 10Y \text{ Yield}$					0.024*** (0.008)	0.025*** (0.008)	0.010 (0.008)	0.010 (0.008)
$\Delta 1M \text{ Yield}$						-0.009 (0.010)	-0.006 (0.009)	-0.006 (0.009)
$\Delta 10Y \text{ Spread}$							0.244*** (0.024)	0.245*** (0.024)
$\Delta 3M \text{ Spread}$								0.021*** (0.008)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	-0.001	0.015	0.014	0.019	0.026	0.026	0.111	0.117

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 36: Post-2020 Japan Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{JPY})$	0.045*** (0.006)	0.040*** (0.006)	0.036*** (0.006)	0.031*** (0.006)	0.030*** (0.006)	0.030*** (0.006)	0.030*** (0.006)	0.030*** (0.006)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.004*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.000** (0.000)	-0.000** (0.000)	-0.000** (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.010** (0.005)	0.010** (0.005)	0.005 (0.005)	0.005 (0.005)
$\Delta 1M \text{ Yield}$						-0.000 (0.005)	0.001 (0.004)	0.001 (0.004)
$\Delta 10Y \text{ Spread}$							0.126*** (0.013)	0.126*** (0.013)
$\Delta 3M \text{ Spread}$								0.008 (0.014)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.027	0.039	0.055	0.057	0.059	0.058	0.097	0.096

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

$\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.008$ pre-2020; ≈ 1.01 post-2020), consistent with small residual gaps and a modest but persistent equity-like loading only in the later period.

Table 37: Pre-2020 Japan Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{JPY})$	-0.002 (0.006)	-0.002 (0.006)	-0.002 (0.006)	-0.003 (0.006)	-0.003 (0.006)	-0.003 (0.006)	-0.003 (0.006)	-0.001 (0.006)
$\Delta \text{Commodity}$		0.003*** (0.001)	0.002*** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)
ΔDollar			-0.004*** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.003** (0.002)	-0.004*** (0.001)	-0.004** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.026** (0.010)	0.027** (0.011)	0.008 (0.010)	0.008 (0.010)
$\Delta 1M \text{ Yield}$						-0.017 (0.013)	-0.013 (0.012)	-0.012 (0.012)
$\Delta 10Y \text{ Spread}$							0.263*** (0.028)	0.265*** (0.028)
$\Delta 3M \text{ Spread}$								0.026*** (0.009)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	-0.001	0.020	0.029	0.030	0.037	0.038	0.148	0.158

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Interpretation. For Japan, the pre-2020 regressions show that as common-shock controls are added, the Litecoin slope stays essentially near zero (from -0.002 in col. 1 to about -0.003 in cols. 4–7, ending at -0.001 in col. 8), while fit improves modestly (Adj. R^2 : $-0.001 \rightarrow 0.158$). This is consistent with a low-comovement narrative: once commodities, dollar, volatility, and rates are accounted for, $\Delta \log(\text{LTC}_{JPY})$ adds little

Table 38: Post-2020 Japan Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{JPY})$	0.019*** (0.005)	0.015*** (0.005)	0.013** (0.005)	0.006 (0.006)	0.006 (0.006)	0.006 (0.006)	0.006 (0.005)	0.006 (0.005)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.000** (0.000)	-0.000** (0.000)
$\Delta 10Y \text{ Yield}$					0.011** (0.005)	0.011** (0.005)	0.006 (0.005)	0.006 (0.005)
$\Delta 1M \text{ Yield}$						-0.000 (0.005)	0.001 (0.005)	0.001 (0.005)
$\Delta 10Y \text{ Spread}$							0.127*** (0.013)	0.127*** (0.013)
$\Delta 3M \text{ Spread}$								0.008 (0.014)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.006	0.021	0.040	0.047	0.049	0.048	0.087	0.087

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

incremental explanatory power for equities. Post-2020, by contrast, the bivariate LTC coefficient is positive and significant (0.019, col. 1) but is largely absorbed by the controls, shrinking to roughly 0.006 and losing significance in cols. 4–8, with Adj. R^2 rising from 0.006 to ~ 0.087 . Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.013$ pre-2020; ≈ 1.024 post-2020), consistent with small residual gaps and only a modest, control-mediated equity-like loading in the later period.

Table 39: Pre-2020 Japan Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{JPY})$	0.030 (0.030)	0.034 (0.030)	0.037 (0.030)	0.030 (0.030)	0.019 (0.031)	0.020 (0.031)	0.027 (0.029)	0.027 (0.029)
$\Delta \text{Commodity}$		0.003*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.003** (0.001)	-0.003* (0.001)	-0.003** (0.001)	-0.003* (0.001)	-0.003** (0.001)	-0.003** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.023** (0.009)	0.024*** (0.009)	0.007 (0.009)	0.007 (0.009)
$\Delta 1M \text{ Yield}$						-0.015 (0.011)	-0.011 (0.011)	-0.010 (0.011)
$\Delta 10Y \text{ Spread}$							0.251*** (0.026)	0.252*** (0.026)
$\Delta 3M \text{ Spread}$								0.021*** (0.008)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	-0.000	0.021	0.025	0.026	0.032	0.033	0.125	0.130

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 40: Post-2020 Japan Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{JPY})$	0.148*** (0.048)	0.168*** (0.047)	0.363*** (0.051)	0.322*** (0.053)	0.331*** (0.059)	0.331*** (0.059)	0.377*** (0.058)	0.378*** (0.058)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.008*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)
ΔVIX				-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					-0.002 (0.005)	-0.002 (0.005)	-0.009* (0.005)	-0.009* (0.005)
$\Delta 1M \text{ Yield}$						-0.001 (0.005)	0.001 (0.004)	0.001 (0.004)
$\Delta 10Y \text{ Spread}$							0.134*** (0.013)	0.134*** (0.013)
$\Delta 3M \text{ Spread}$								0.009 (0.014)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.004	0.023	0.060	0.063	0.062	0.062	0.105	0.105

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For Japan, the pre-2020 regressions show a near-neutral Tether slope that is imprecisely estimated and not robust to controls (from 0.030 in col. 1 to ≈ 0.019 – 0.027 in cols. 5–8, all insignificant), while fit edges up only modestly as common shocks are added (Adj. R^2 : $-0.000 \rightarrow 0.130$). Post-2020, by contrast, the bivariate USDT coefficient is positive and significant (0.148, col. 1) and remains large after controls (≈ 0.32 – 0.38 in cols. 4–8), with a small improvement in fit (Adj. R^2 : $0.004 \rightarrow 0.105$). This persistence suggests that in a JPY setting—where FX dynamics and safe-haven flows are more pronounced—USDT loads on the equity leg through an FX/flow channel not fully absorbed by our standard commodity–dollar–volatility–rate controls. Using the square-root variance ratio $\sqrt{\rho}$, dispersion is elevated pre-2020 and substantial post-2020 ($\sqrt{\rho} \approx 1.295$ pre; ≈ 2.711 post), consistent with peg mechanics and JPY/USD microstructure; thus, while the fully controlled post-2020 slope is high, part of that strength may reflect peg-related noise rather than pure equity co-movement.

Interpretation. For Japan, the pre-2020 regressions show a small and statistically weak Ripple slope that drifts toward zero as common shocks are added (from 0.006 in col. 1 to ≈ 0.003 in cols. 7–8, not significant), while fit rises from essentially none to modest (Adj. R^2 : $-0.000 \rightarrow 0.158$). Post-2020, the bivariate XRP coefficient is significant but small (0.011, col. 1) and is absorbed once controls enter (≈ 0.003 in cols. 4–8, not significant), with only a slight improvement in fit (Adj. R^2 : $0.002 \rightarrow 0.086$). Across both windows, the control block—especially the dollar, VIX, and term-structure proxies—does the explanatory heavy lifting (e.g., large positive loadings on the long–short spread in cols. 7–8), leaving XRP’s incremental equity co-movement negligible. Using the square-root variance ratio $\sqrt{\rho}$, both periods lie close to one ($\sqrt{\rho} \approx 1.010$ pre-2020; ≈ 1.019 post-2020), consistent with residual gaps that are small relative to the idiosyncratic crypto leg and a near-neutral XRP–equity relation in JPY terms.

2.5 China

Interpretation. For China, the pre-2020 regressions show that as common-shock controls are added, the Bitcoin slope contracts from a small positive (0.004 in col. 1) to a small and weak effect (≈ 0.003 in cols. 4–8), while fit rises (Adj. R^2 : $0.000 \rightarrow 0.031$). This pattern is consistent with a “hedge/low comovement” narrative before 2020: once broad macro factors (commodities, dollar, volatility, rates) are accounted for, $\Delta \log(\text{BTC}_{CNY})$ contributes little incremental explanatory power for equities. Post-2020, by contrast, the BTC coefficient is large and highly significant in the bivariate specification (0.020 in col. 1) and attenuates

Table 41: Pre-2020 Japan Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{JPY})$	0.006 (0.007)	0.006 (0.007)	0.005 (0.007)	0.005 (0.007)	0.004 (0.007)	0.004 (0.007)	0.003 (0.006)	0.003 (0.006)
$\Delta \text{Commodity}$		0.003*** (0.001)	0.002*** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)
ΔDollar			-0.004*** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.003** (0.002)	-0.004*** (0.001)	-0.004** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.025** (0.010)	0.027** (0.011)	0.008 (0.010)	0.008 (0.010)
$\Delta 1M \text{ Yield}$						-0.017 (0.013)	-0.013 (0.012)	-0.012 (0.012)
$\Delta 10Y \text{ Spread}$							0.263*** (0.028)	0.264*** (0.028)
$\Delta 3M \text{ Spread}$								0.026*** (0.009)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	-0.000	0.021	0.029	0.031	0.037	0.038	0.148	0.158

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 42: Post-2020 Japan Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{JPY})$	0.011** (0.005)	0.008* (0.005)	0.007 (0.005)	0.003 (0.005)	0.003 (0.005)	0.003 (0.005)	0.003 (0.005)	0.003 (0.005)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
$\Delta 10Y \text{ Yield}$					0.011** (0.005)	0.011** (0.005)	0.006 (0.005)	0.006 (0.005)
$\Delta 1M \text{ Yield}$						-0.000 (0.005)	0.001 (0.005)	0.001 (0.005)
$\Delta 10Y \text{ Spread}$							0.127*** (0.013)	0.127*** (0.013)
$\Delta 3M \text{ Spread}$								0.008 (0.014)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.002	0.019	0.038	0.046	0.048	0.048	0.087	0.086

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 43: Pre-2020 Japan Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{CNY})$	0.004 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.002** (0.001)	-0.001* (0.001)	-0.001* (0.001)	-0.001* (0.001)	-0.001* (0.001)	-0.001* (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					-0.000 (0.005)	-0.000 (0.005)	-0.001 (0.005)	-0.001 (0.005)
$\Delta 1\text{M Yield}$						-0.000 (0.010)	-0.000 (0.010)	-0.000 (0.010)
$\Delta 10\text{Y Spread}$							0.028*** (0.009)	0.033*** (0.009)
$\Delta 1\text{Y Spread}$								-0.012** (0.006)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	0.000	0.023	0.024	0.027	0.027	0.027	0.030	0.031

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 44: Post-2020 Japan Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{CNY})$	0.020*** (0.006)	0.014** (0.006)	0.008 (0.006)	0.006 (0.006)	0.006 (0.006)	0.006 (0.006)	0.005 (0.006)	0.005 (0.006)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.008** (0.004)	0.008** (0.004)	0.006 (0.004)	0.006 (0.004)
$\Delta 1\text{M Yield}$						-0.001 (0.004)	-0.002 (0.003)	-0.002 (0.003)
$\Delta 10\text{Y Spread}$							0.094*** (0.011)	0.105*** (0.011)
$\Delta 1\text{Y Spread}$								-0.026*** (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.005	0.026	0.055	0.055	0.056	0.056	0.089	0.094

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

after controls to a small effect (≈ 0.005 in cols. 7–8), with Adj. R^2 improving from 0.005 to ~ 0.094 . This aligns with the paper’s model in which crypto–FX–equity linkages induce stronger equity-like comovement after 2020, though much of the raw comovement is absorbed by common shocks. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.016$ pre-2020; ≈ 1.035 post-2020), indicating a slightly wider residual gap post-2020 and a shift from hedge-like behavior toward modest positive co-movement.

Table 45: Pre-2020 Japan Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{CNY})$	0.003 (0.004)	0.002 (0.004)	0.001 (0.004)	0.001 (0.004)	0.001 (0.004)	0.001 (0.004)	0.001 (0.004)	0.001 (0.004)
$\Delta \text{Commodity}$		0.004*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)
ΔDollar			-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)
ΔVIX				0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.022*** (0.008)	0.022*** (0.008)	0.017** (0.008)	0.017** (0.008)
$\Delta 1\text{M Yield}$						-0.010 (0.010)	-0.009 (0.010)	-0.009 (0.010)
$\Delta 10\text{Y Spread}$							0.074*** (0.013)	0.074*** (0.013)
$\Delta 1\text{Y Spread}$								-0.001 (0.010)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	-0.001	0.047	0.052	0.052	0.057	0.057	0.084	0.083

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 46: Post-2020 Japan Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{CNY})$	0.022*** (0.005)	0.017*** (0.005)	0.012*** (0.005)	0.012** (0.005)	0.012** (0.005)	0.012** (0.005)	0.011** (0.005)	0.011** (0.005)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.008** (0.004)	0.008** (0.004)	0.006 (0.004)	0.006 (0.004)
$\Delta 1\text{M Yield}$						-0.001 (0.004)	-0.002 (0.003)	-0.002 (0.003)
$\Delta 10\text{Y Spread}$							0.094*** (0.011)	0.105*** (0.011)
$\Delta 1\text{Y Spread}$								-0.026*** (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.010	0.030	0.057	0.057	0.059	0.058	0.091	0.097

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For China, the pre-2020 regressions show that as common-shock controls are added, the Ethereum slope contracts from a small and insignificant 0.003 in col. 1 to a near-zero and imprecise value

(≈ 0.001 in cols. 4–8), while fit improves moderately (Adj. R^2 : $-0.001 \rightarrow 0.083$). This pattern is consistent with a “hedge/low comovement” narrative before 2020: once broad macro factors (commodities, dollar, volatility, rates) are accounted for, $\Delta \log(\text{ETH}_{CNY})$ adds little explanatory power for equities. Post-2020, by contrast, the ETH coefficient is large and highly significant in the bivariate regression (0.022 in col. 1) and remains economically meaningful even after full controls (≈ 0.011 in cols. 7–8), with model fit improving from 0.010 to ~ 0.097 . This aligns with the paper’s model in which crypto–FX–equity linkages strengthen after 2020, suggesting increased equity-like comovement in the Chinese market. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.008$ pre-2020; ≈ 1.012 post-2020), indicating only a slight narrowing of the residual gap post-2020 and a mild shift from hedge-like behavior toward sustained positive co-movement.

Table 47: Pre-2020 Japan Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{CNY})$	0.014* (0.007)	0.014** (0.007)	0.014** (0.007)	0.014** (0.007)	0.014** (0.007)	0.015** (0.007)	0.013* (0.007)	0.013* (0.007)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)
ΔDollar			-0.004** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.003** (0.002)	-0.003** (0.002)
ΔVIX				-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.027** (0.012)	0.028** (0.012)	0.013 (0.012)	0.013 (0.012)
$\Delta 1\text{M Yield}$						-0.019 (0.015)	-0.018 (0.015)	-0.018 (0.015)
$\Delta 10\text{Y Spread}$							0.126*** (0.019)	0.126*** (0.020)
$\Delta 1\text{Y Spread}$								-0.000 (0.013)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.004	0.066	0.072	0.071	0.077	0.078	0.131	0.129

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For China, the pre-2020 regressions show that as common-shock controls are added, the Litecoin slope remains small but persistently positive (from 0.014 in col. 1 to ≈ 0.013 in cols. 7–8), while fit rises (Adj. R^2 : $0.004 \rightarrow 0.129$). This is consistent with a modest equity-like signal that survives after accounting for broad macro factors (commodities, dollar, volatility, rates, and term spreads), albeit at a small magnitude. Post-2020, the bivariate LTC coefficient is slightly larger (0.016, col. 1) but compresses once the same controls are included (to ≈ 0.009 in cols. 4–6 and ~ 0.008 in cols. 7–8), with Adj. R^2 improving from 0.007 to ~ 0.096 . Taken together, the tables indicate that most apparent comovement in the later period is absorbed by common shocks, leaving only a modest incremental role for $\Delta \log(\text{LTC}_{CNY})$ once controls are present. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.003$ pre-2020; ≈ 1.009 post-2020), indicating similar residual parity across periods and reinforcing that LTC’s equity-like loading in China is positive but small after controls.

Interpretation. For China, the pre-2020 regressions show that as common-shock controls are added, the Tether slope moves from a modest negative (-0.070 in col. 1) toward a small, imprecise value (≈ -0.015 in cols. 7–8), while fit rises (Adj. R^2 : $0.004 \rightarrow 0.122$). This is consistent with a peg-neutral/low-comovement narrative: once broad macro factors (commodities, dollar, volatility, rates, term spreads) are netted out, $\Delta \log(\text{USDT}_{CNY})$ adds little incremental explanatory power for equities. Post-2020, by contrast, the bivariate USDT coefficient is sharply negative (-0.741 , col. 1) and remains sizable after controls ($\approx$

Table 48: Post-2020 Japan Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{CNY})$	0.016*** (0.004)	0.013*** (0.004)	0.010** (0.004)	0.009** (0.004)	0.009** (0.004)	0.009** (0.004)	0.008** (0.004)	0.008** (0.004)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.008** (0.004)	0.008** (0.004)	0.006* (0.004)	0.006 (0.004)
$\Delta 1\text{M Yield}$						-0.001 (0.004)	-0.002 (0.003)	-0.002 (0.003)
$\Delta 10\text{Y Spread}$							0.093*** (0.011)	0.105*** (0.011)
$\Delta 1\text{Y Spread}$								-0.026*** (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.007	0.028	0.057	0.056	0.058	0.058	0.090	0.096

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 49: Pre-2020 Japan Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{CNY})$	-0.070** (0.034)	-0.044 (0.034)	-0.034 (0.034)	-0.034 (0.034)	-0.031 (0.034)	-0.030 (0.034)	-0.015 (0.033)	-0.015 (0.033)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)
ΔDollar			-0.004** (0.001)	-0.004** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
ΔVIX				0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
$\Delta 10\text{Y Yield}$					0.026*** (0.010)	0.027*** (0.010)	0.016* (0.010)	0.016* (0.010)
$\Delta 1\text{M Yield}$						-0.011 (0.012)	-0.012 (0.012)	-0.012 (0.012)
$\Delta 10\text{Y Spread}$							0.120*** (0.016)	0.121*** (0.016)
$\Delta 1\text{Y Spread}$								-0.004 (0.011)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	0.004	0.058	0.064	0.063	0.069	0.069	0.123	0.122

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 50: Post-2020 Japan Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{CNY})$	-0.741*** (0.075)	-0.647*** (0.077)	-0.447*** (0.086)	-0.449*** (0.086)	-0.455*** (0.086)	-0.456*** (0.086)	-0.389*** (0.085)	-0.385*** (0.085)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.003*** (0.001)	-0.003*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.009** (0.004)	0.009** (0.004)	0.007* (0.004)	0.006* (0.004)
$\Delta 1M \text{ Yield}$						-0.001 (0.003)	-0.002 (0.003)	-0.002 (0.003)
$\Delta 10Y \text{ Spread}$							0.089*** (0.011)	0.100*** (0.011)
$\Delta 1Y \text{ Spread}$								-0.026*** (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.044	0.055	0.066	0.066	0.069	0.068	0.097	0.103

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

−0.389 in cols. 7–8), with Adj. R^2 improving from 0.044 to ~ 0.103 . Using the square-root variance ratio $\sqrt{\rho}$, residual parity loosens markedly across windows ($\sqrt{\rho} \approx 1.401$ pre-2020; ≈ 4.116 post-2020), indicating much larger residual dispersion in the later period—consistent with peg mechanics and FX conversion noise rather than genuine equity linkage, and reinforcing that controlled estimates, not bivariate loadings, are most informative here.

Table 51: Pre-2020 Japan Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{CNY})$	0.014* (0.008)	0.015* (0.008)	0.015* (0.008)	0.014* (0.008)	0.014* (0.008)	0.015* (0.008)	0.015** (0.007)	0.015** (0.007)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)
ΔDollar			-0.004** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.003** (0.002)	-0.003** (0.002)
ΔVIX				-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.027** (0.012)	0.028** (0.012)	0.013 (0.012)	0.013 (0.012)
$\Delta 1M \text{ Yield}$						-0.019 (0.015)	-0.018 (0.015)	-0.018 (0.015)
$\Delta 10Y \text{ Spread}$							0.127*** (0.019)	0.127*** (0.020)
$\Delta 1Y \text{ Spread}$								0.000 (0.013)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.003	0.065	0.072	0.071	0.076	0.077	0.131	0.130

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 52: Post-2020 Japan Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{CNY})$	0.012*** (0.004)	0.010*** (0.004)	0.008** (0.004)	0.008** (0.004)	0.008** (0.004)	0.008** (0.004)	0.008** (0.004)	0.008** (0.004)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
$\Delta 10Y \text{ Yield}$					0.008** (0.004)	0.008** (0.004)	0.006* (0.004)	0.006 (0.004)
$\Delta 1M \text{ Yield}$						-0.001 (0.004)	-0.002 (0.003)	-0.002 (0.003)
$\Delta 10Y \text{ Spread}$							0.094*** (0.011)	0.105*** (0.011)
$\Delta 1Y \text{ Spread}$								-0.026*** (0.007)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.005	0.027	0.057	0.056	0.058	0.058	0.090	0.096

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For China, the pre-2020 regressions indicate *near-neutral/low comovement*: the XRP slope is small in the bivariate case (0.014, col. 1) and remains of similar order after controls (≈ 0.015 in cols. 7–8), while fit rises (Adj. R^2 : 0.003 $\rightarrow$ 0.130). Post-2020, any apparent bivariate loading is likewise small (0.012, col. 1) and contracts once common shocks are netted out (≈ 0.008 in cols. 4–8), with Adj. R^2 improving from 0.005 to ~ 0.096 . Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.004$ pre-2020; ≈ 1.005 post-2020), indicating similar residual parity across periods and reinforcing a low-comovement narrative in which $\Delta \log(\text{XRP}_{CNY})$ contributes little incremental explanatory power after controls.

2.6 Brazil

Interpretation. For Brazil, the pre-2020 regressions show that as common-shock controls are added, the Bitcoin slope moves from a small negative in the bivariate specification (-0.005 in col. 1) to essentially zero with full controls (cols. 7–8), while fit rises (Adj. R^2 : 0.000 $\rightarrow$ 0.322). This pattern is consistent with a “hedge/low comovement” narrative before 2020: once broad macro factors (commodities, dollar, volatility, rates, and term spreads) are accounted for, $\Delta \log(\text{BTC}_{BRL})$ contributes little incremental explanatory power for equities. Post-2020, by contrast, the BTC coefficient is large and positive in the bivariate regression (0.048, col. 1) but is absorbed by controls—turning modestly negative mid-spec (about -0.016 in cols. 4–6) and ending near zero with spreads (cols. 7–8)—with Adj. R^2 improving from 0.013 to ~ 0.444 . Using the square-root variance ratio $\sqrt{\rho}$, the pre-2020 window lies near unity ($\sqrt{\rho} \approx 1.014$), while the post-2020 window is moderately above one (≈ 1.051), indicating a looser parity (larger residual gap) later and reinforcing that much of the raw post-2020 comovement reflects common shocks rather than a robust standalone BTC–equity linkage.

Interpretation. For Brazil, the pre-2020 regressions show that as common-shock controls are added, the Ethereum slope turns more negative (from -0.009 in col. 1 to about -0.014 in cols. 4–6) before attenuating toward zero once term spreads enter (to ≈ -0.005 to -0.006 in cols. 7–8), while fit rises (Adj. R^2 : 0.002 $\rightarrow$ 0.416). This is consistent with a pre-2020 “hedge/offset” pattern in which broad macro factors (commodities, dollar, volatility, rates) absorb most equity variation and $\Delta \log(\text{ETH}_{BRL})$ adds, at most, a small negative contribution that weakens when spread structure is controlled for. Post-2020, by contrast, the bivariate ETH coefficient is large and positive (0.047, col. 1) but is largely absorbed by controls—shrinking to an insignificant

Table 53: Pre-2020 Brazil Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{BRL})$	-0.005 (0.004)	-0.004 (0.003)	-0.002 (0.003)	-0.004 (0.003)	-0.004 (0.003)	-0.004 (0.003)	0.001 (0.003)	0.001 (0.003)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
ΔVIX				-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
$\Delta 10\text{Y Yield}$					0.012** (0.005)	0.012** (0.005)	0.021*** (0.004)	0.022*** (0.004)
$\Delta 1\text{M Yield}$						0.002 (0.009)	-0.001 (0.009)	-0.004 (0.009)
$\Delta 10\text{Y Spread}$							-0.029*** (0.002)	-0.025*** (0.002)
$\Delta 1\text{Y Spread}$								-0.015*** (0.003)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	0.000	0.102	0.138	0.239	0.241	0.240	0.316	0.322

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 54: Post-2020 Brazil Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{BRL})$	0.048*** (0.009)	0.040*** (0.009)	0.032*** (0.009)	-0.015* (0.008)	-0.016** (0.008)	-0.016** (0.008)	0.000 (0.007)	-0.000 (0.007)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.002*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
ΔVIX				-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
$\Delta 10\text{Y Yield}$					0.021*** (0.005)	0.020*** (0.005)	0.033*** (0.004)	0.032*** (0.004)
$\Delta 1\text{M Yield}$						0.006 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.039*** (0.002)	-0.034*** (0.002)
$\Delta 1\text{Y Spread}$								-0.016*** (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.013	0.064	0.113	0.301	0.307	0.308	0.435	0.444

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 55: Pre-2020 Ethereum Brazil Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{BRL})$	-0.009 (0.005)	-0.009* (0.005)	-0.011** (0.005)	-0.014*** (0.005)	-0.014*** (0.005)	-0.014*** (0.005)	-0.005 (0.004)	-0.006 (0.004)
$\Delta \text{Commodity}$		0.006*** (0.001)	0.005*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.003*** (0.001)	0.003*** (0.001)
ΔDollar			-0.007*** (0.001)	-0.006*** (0.001)	-0.007*** (0.001)	-0.006*** (0.001)	-0.003** (0.001)	-0.002** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.016* (0.009)	0.016* (0.009)	0.024*** (0.008)	0.023*** (0.008)
$\Delta 1\text{M Yield}$						-0.007 (0.012)	-0.013 (0.010)	-0.013 (0.010)
$\Delta 10\text{Y Spread}$							-0.048*** (0.002)	-0.035*** (0.003)
$\Delta 1\text{Y Spread}$								-0.026*** (0.004)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	0.002	0.077	0.102	0.173	0.174	0.174	0.396	0.416

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 56: Post-2020 Ethereum Brazil Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{BRL})$	0.047*** (0.007)	0.040*** (0.007)	0.032*** (0.007)	-0.008 (0.006)	-0.008 (0.006)	-0.008 (0.006)	0.002 (0.005)	0.001 (0.005)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.002*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
ΔVIX				-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
$\Delta 10\text{Y Yield}$					0.021*** (0.005)	0.020*** (0.005)	0.033*** (0.004)	0.032*** (0.004)
$\Delta 1\text{M Yield}$						0.006 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.039*** (0.002)	-0.034*** (0.002)
$\Delta 1\text{Y Spread}$								-0.016*** (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.022	0.070	0.117	0.301	0.307	0.307	0.435	0.444

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

≈ -0.008 with core controls (cols. 4–6) and settling near zero once spreads are included (cols. 7–8)—with Adj. R^2 improving from 0.022 to ~ 0.444 . Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near (but above) unity ($\sqrt{\rho} \approx 1.015$ pre-2020; ≈ 1.029 post-2020), indicating a somewhat looser parity later (larger residual gap) and reinforcing that much of the raw post-2020 comovement reflects common shocks rather than a robust standalone ETH–equity linkage.

Table 57: Pre-2020 Brazil Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{BRL})$	-0.003 (0.008)	-0.001 (0.008)	-0.001 (0.007)	-0.002 (0.007)	-0.002 (0.007)	-0.002 (0.007)	0.005 (0.006)	0.006 (0.006)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.008*** (0.002)	-0.006*** (0.002)	-0.006*** (0.002)	-0.006*** (0.002)	-0.003** (0.002)	-0.003* (0.002)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.011 (0.012)	0.013 (0.012)	0.021* (0.011)	0.018* (0.010)
$\Delta 1\text{M Yield}$						-0.017 (0.015)	-0.017 (0.013)	-0.017 (0.013)
$\Delta 10\text{Y Spread}$							-0.048*** (0.003)	-0.036*** (0.004)
$\Delta 1\text{Y Spread}$								-0.028*** (0.005)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	-0.001	0.061	0.087	0.160	0.160	0.161	0.355	0.382

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 58: Post-2020 Brazil Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{BRL})$	0.031*** (0.006)	0.026*** (0.006)	0.022*** (0.006)	-0.008 (0.005)	-0.008 (0.005)	-0.008 (0.005)	0.001 (0.005)	0.000 (0.005)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.002*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
ΔVIX				-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
$\Delta 10\text{Y Yield}$					0.020*** (0.005)	0.020*** (0.005)	0.033*** (0.004)	0.032*** (0.004)
$\Delta 1\text{M Yield}$						0.006 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.039*** (0.002)	-0.034*** (0.002)
$\Delta 1\text{Y Spread}$								-0.016*** (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.012	0.063	0.113	0.301	0.307	0.307	0.435	0.444

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For Brazil, the pre-2020 regressions show weak, imprecise comovement: the Litecoin slope is near zero in the bivariate specification (-0.003 , col. 1) and remains economically small once common-shock controls are added (drifting toward ≈ 0 by cols. 4–6 and to 0.005 – 0.006 with term spreads in cols. 7–8), while fit rises sharply (Adj. R^2 : $-0.001 \rightarrow 0.382$). This is consistent with a pre-2020 “hedge/low comovement” pattern in which broad macro factors (commodities, dollar, volatility, rates, term structure) explain most equity variation, leaving $\Delta \log(\text{LTC}_{BRL})$ with little incremental power. Post-2020, the bivariate LTC coefficient is larger (0.031 , col. 1) but collapses once controls are included (to about -0.008 in cols. 4–6 and ≈ 0.000 – 0.001 in cols. 7–8), even as Adj. R^2 improves from 0.012 to ~ 0.444 . Taken together, the later-period comovement is largely absorbed by common shocks, leaving little robust role for $\Delta \log(\text{LTC}_{BRL})$ after controls. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie close to one ($\sqrt{\rho} \approx 1.007$ pre-2020; ≈ 1.023 post-2020), implying a slightly looser parity (larger residual gap) in the later window, consistent with the small and non-robust incremental loading of LTC on Brazilian equities after 2020.

Table 59: Pre-2020 Brazil Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{BRL})$	-0.264*** (0.029)	-0.227*** (0.029)	-0.211*** (0.029)	-0.195*** (0.028)	-0.197*** (0.028)	-0.196*** (0.028)	-0.057** (0.027)	-0.049* (0.026)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.006*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.003** (0.001)	-0.003** (0.001)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.021** (0.010)	0.022** (0.010)	0.028*** (0.009)	0.026*** (0.009)
$\Delta 1\text{M Yield}$						-0.008 (0.013)	-0.010 (0.011)	-0.010 (0.011)
$\Delta 10\text{Y Spread}$							-0.046*** (0.003)	-0.036*** (0.004)
$\Delta 1\text{Y Spread}$								-0.027*** (0.005)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	0.085	0.129	0.145	0.220	0.223	0.223	0.366	0.386

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For Brazil, Tether loads *negatively* and persistently on equities in both windows, consistent with a currency-risk channel: when $\Delta \log(\text{USDT}_{BRL})$ rises (BRL weakens), equity returns fall. Pre-2020, the bivariate slope is large and negative (-0.264 , col. 1) and, while attenuating with controls, remains negative and economically meaningful (about -0.05 in cols. 7–8), as fit improves sharply (Adj. R^2 : $0.085 \rightarrow 0.386$). Post-2020, the pattern strengthens: the bivariate coefficient is very large (-0.783 , col. 1) and stays strongly negative with full controls (≈ -0.37 in cols. 7–8), with fit rising further (Adj. R^2 : $0.253 \rightarrow 0.482$). Using the square-root variance ratio $\sqrt{\rho}$, residual parity is notably looser post-2020 ($\sqrt{\rho} \approx 1.308$ pre-2020; ≈ 1.924 post-2020), in line with a more pronounced currency–equity linkage in the later period.

Interpretation. For Brazil, the pre-2020 regressions show weak, imprecise comovement: the Ripple slope is essentially zero in the bivariate specification (-0.000 , col. 1) and remains economically small once common-shock controls are added (hovering near 0 and reaching 0.003 with term spreads in cols. 7–8), while fit rises sharply (Adj. R^2 : $-0.001 \rightarrow 0.381$). This is consistent with a pre-2020 “hedge/low comovement” pattern in which broad macro factors (commodities, dollar, volatility, rates, term structure) explain most equity variation, leaving $\Delta \log(\text{XRP}_{BRL})$ with little incremental power. Post-2020, the bivariate XRP coefficient is larger (0.016 , col. 1) but largely collapses once controls are included (to about -0.005 in cols. 4–6 and ≈ 0.002 in cols. 7–8), even as Adj. R^2 improves from 0.004 to ~ 0.444 . Taken together, the later-period comovement is absorbed by common shocks, leaving little robust role for $\Delta \log(\text{XRP}_{BRL})$ after controls. Us-

Table 60: Post-2020 Brazil Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{BRL})$	-0.783*** (0.029)	-0.738*** (0.030)	-0.688*** (0.032)	-0.557*** (0.029)	-0.560*** (0.029)	-0.560*** (0.029)	-0.377*** (0.030)	-0.367*** (0.030)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001* (0.000)	0.000 (0.000)	0.000 (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.004*** (0.001)	-0.003*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
ΔVIX				-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.022*** (0.004)	0.021*** (0.004)	0.031*** (0.004)	0.031*** (0.004)
$\Delta 1\text{M Yield}$						0.006 (0.004)	0.003 (0.004)	0.004 (0.004)
$\Delta 10\text{Y Spread}$							-0.030*** (0.002)	-0.026*** (0.002)
$\Delta 1\text{Y Spread}$								-0.014*** (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.253	0.262	0.271	0.403	0.410	0.410	0.475	0.482

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 61: Pre-2020 Brazil Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{BRL})$	-0.000 (0.008)	0.002 (0.008)	0.002 (0.008)	0.000 (0.008)	0.000 (0.008)	0.001 (0.008)	0.003 (0.007)	0.003 (0.007)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.004*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.008*** (0.002)	-0.006*** (0.002)	-0.006*** (0.002)	-0.006*** (0.002)	-0.003** (0.002)	-0.003* (0.002)
ΔVIX				-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
$\Delta 10\text{Y Yield}$					0.011 (0.012)	0.013 (0.012)	0.021* (0.011)	0.018* (0.010)
$\Delta 1\text{M Yield}$						-0.017 (0.015)	-0.017 (0.013)	-0.017 (0.013)
$\Delta 10\text{Y Spread}$							-0.048*** (0.003)	-0.036*** (0.004)
$\Delta 1\text{Y Spread}$								-0.028*** (0.005)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	-0.001	0.061	0.087	0.160	0.160	0.161	0.355	0.381

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 62: Post-2020 Brazil Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{BRL})$	0.016*** (0.005)	0.013** (0.005)	0.012** (0.005)	-0.005 (0.005)	-0.005 (0.005)	-0.005 (0.005)	0.002 (0.004)	0.002 (0.004)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.002*** (0.000)
ΔDollar			-0.009*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
ΔVIX				-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
$\Delta 10\text{Y Yield}$					0.020*** (0.005)	0.020*** (0.005)	0.033*** (0.004)	0.032*** (0.004)
$\Delta 1\text{M Yield}$						0.006 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.039*** (0.002)	-0.034*** (0.002)
$\Delta 1\text{Y Spread}$								-0.016*** (0.003)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.004	0.057	0.110	0.300	0.306	0.307	0.435	0.444

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

ing the square-root variance ratio $\sqrt{\rho}$, both windows lie near one ($\sqrt{\rho} \approx 1.012$ pre-2020; ≈ 1.015 post-2020), implying a slightly looser parity (larger residual gap) in the later window.

2.7 India

Table 63: Pre-2020 India Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{INR})$	-0.008*** (0.002)	-0.008*** (0.002)	-0.008*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.003*** (0.001)	-0.002*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.006* (0.003)	0.006 (0.003)	0.006* (0.003)	0.006* (0.003)
$\Delta 1\text{M Yield}$						0.007 (0.007)	0.007 (0.007)	0.007 (0.007)
$\Delta 10\text{Y Spread}$							-0.025*** (0.003)	-0.022*** (0.003)
$\Delta 2\text{Y Spread}$								-0.005* (0.003)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	0.003	0.041	0.049	0.069	0.070	0.070	0.087	0.088

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For India, the pre-2020 regressions show a persistently negative and statistically precise BTC slope that slightly strengthens as common-shock controls are added (-0.008 in col. 1 to ≈ -0.009

Table 64: Post-2020 India Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{INR})$	0.028*** (0.007)	0.023*** (0.007)	0.010 (0.007)	-0.008 (0.007)	-0.008 (0.007)	-0.008 (0.007)	-0.008 (0.007)	-0.007 (0.007)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
ΔDollar			-0.009*** (0.001)	-0.008*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.017*** (0.004)	0.017*** (0.004)	0.018*** (0.004)	0.018*** (0.004)
$\Delta 1\text{M Yield}$						0.002 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.019*** (0.007)	-0.012 (0.007)
$\Delta 2\text{Y Spread}$								-0.012** (0.005)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.008	0.018	0.101	0.132	0.140	0.140	0.142	0.144

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

in cols. 4–8), while fit rises (Adj. R^2 : 0.003 $\rightarrow$ 0.088). This is consistent with a pre-2020 “hedge/low comovement” profile in which broad macro factors (commodities, dollar, volatility, rates, and term spreads) account for most equity variation, leaving $\Delta \log(\text{BTC}_{INR})$ with a small offsetting sign and limited incremental power. Post-2020, the BTC coefficient is positive and significant in the bivariate regression (0.028, col. 1) but collapses toward a small, slightly negative and statistically weak value once controls are included (≈ -0.008 to -0.007 in cols. 4–8), even as Adj. R^2 improves from 0.008 to ~ 0.144 . Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.018$ pre-2020; ≈ 1.053 post-2020), implying a modest loosening of parity in the later window (a larger residual gap) and reinforcing that any raw post-2020 comovement is largely absorbed by common shocks once controls are present.

Interpretation. For India, the pre-2020 regressions show that as common-shock controls are added, the Ethereum slope moves from a small negative in the bivariate specification (-0.007 in col. 1) to a slightly larger and statistically robust negative (about -0.009 in cols. 4–8), while fit improves (Adj. R^2 : 0.003 $\rightarrow$ 0.031). This is consistent with a pre-2020 “hedge/low comovement” narrative in which broad macro factors (commodities, dollar, volatility, rates) absorb most equity variation, leaving $\Delta \log(\text{ETH}_{INR})$ with little incremental explanatory power and a weak offsetting sign once controls are included. Post-2020, the ETH coefficient is large and highly significant in the bivariate regression (0.033, col. 1) but attenuates to a small and statistically negligible ≈ 0.006 after controls (cols. 4–8), even as Adj. R^2 rises from 0.020 to ~ 0.144 . Taken together, the later-period comovement is largely absorbed by common shocks, leaving a modest residual loading. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie close to unity ($\sqrt{\rho} \approx 1.015$ pre-2020; ≈ 1.020 post-2020), indicating only a slight widening of the residual gap in the later period and reinforcing the transition from hedge-like behavior to modest positive co-movement.

Interpretation. For India, the pre-2020 regressions show weak, slightly negative comovement: the Litecoin slope is small and stable at -0.006 across all specifications (cols. 1–8, SE ≈ 0.005), while fit rises only modestly as common-shock controls (commodities, dollar, volatility, rates, and term spreads) are added (Adj. R^2 : 0.000 $\rightarrow$ 0.038). This is consistent with a pre-2020 “hedge/low comovement” pattern in which broad macro factors explain most equity variation, leaving $\Delta \log(\text{LTC}_{INR})$ with little incremental power. Post-2020, the bivariate LTC coefficient is positive and significant (0.014, col. 1) but collapses once controls are included (to about -0.006 by cols. 4–8), even as model fit improves substantially (Adj. R^2 : 0.004 $\rightarrow$ 0.144). Taken together, apparent later-period comovement is largely absorbed by common shocks, leaving a negligible

Table 65: Pre-2020 India Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{INR})$	-0.007** (0.003)	-0.008** (0.003)	-0.008** (0.003)	-0.009*** (0.003)	-0.009*** (0.003)	-0.009*** (0.003)	-0.009*** (0.003)	-0.009*** (0.003)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001** (0.000)	0.001* (0.000)	0.001** (0.000)	0.001** (0.000)	0.001** (0.000)	0.001** (0.000)
ΔDollar			-0.003*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)
ΔVIX				-0.000*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
$\Delta 10\text{Y Yield}$					-0.004 (0.006)	-0.004 (0.006)	-0.003 (0.006)	-0.003 (0.006)
$\Delta 1\text{M Yield}$						0.002 (0.008)	0.002 (0.008)	0.002 (0.008)
$\Delta 10\text{Y Spread}$							-0.004 (0.005)	-0.002 (0.005)
$\Delta 2\text{Y Spread}$								-0.006 (0.006)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	0.003	0.017	0.025	0.032	0.032	0.031	0.031	0.031

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 66: Post-2020 India Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{INR})$	0.033*** (0.005)	0.030*** (0.005)	0.019*** (0.005)	0.006 (0.005)	0.006 (0.005)	0.006 (0.005)	0.006 (0.005)	0.006 (0.005)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
ΔDollar			-0.009*** (0.001)	-0.008*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.008*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.017*** (0.004)	0.017*** (0.004)	0.018*** (0.004)	0.018*** (0.004)
$\Delta 1\text{M Yield}$						0.003 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.018*** (0.007)	-0.011 (0.007)
$\Delta 2\text{Y Spread}$								-0.012** (0.005)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.020	0.029	0.107	0.133	0.140	0.140	0.142	0.144

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 67: Pre-2020 India Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{INR})$	-0.006 (0.005)	-0.006 (0.005)	-0.006 (0.005)	-0.006 (0.005)	-0.006 (0.005)	-0.006 (0.005)	-0.006 (0.005)	-0.006 (0.005)
$\Delta \text{Commodity}$		0.002*** (0.001)	0.001 (0.001)	0.000 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
ΔDollar			-0.005*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
ΔVIX				-0.001** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					-0.013 (0.009)	-0.014 (0.009)	-0.013 (0.009)	-0.012 (0.009)
$\Delta 1\text{M Yield}$						0.008 (0.011)	0.008 (0.011)	0.008 (0.011)
$\Delta 10\text{Y Spread}$							-0.006 (0.007)	-0.003 (0.007)
$\Delta 2\text{Y Spread}$								-0.009 (0.008)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.000	0.012	0.030	0.036	0.038	0.037	0.037	0.037

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 68: Post-2020 India Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{INR})$	0.014*** (0.005)	0.011** (0.005)	0.005 (0.004)	-0.006 (0.004)	-0.006 (0.004)	-0.006 (0.004)	-0.005 (0.004)	-0.005 (0.004)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
ΔDollar			-0.009*** (0.001)	-0.008*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.017*** (0.004)	0.017*** (0.004)	0.018*** (0.004)	0.018*** (0.004)
$\Delta 1\text{M Yield}$						0.003 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.018*** (0.007)	-0.012 (0.007)
$\Delta 2\text{Y Spread}$								-0.012** (0.005)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.004	0.016	0.101	0.133	0.140	0.140	0.142	0.144

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

residual loading for $\Delta \log(\text{LTC}_{INR})$. Using the square-root variance ratio $\sqrt{\rho}$, both windows sit close to one ($\sqrt{\rho} \approx 1.015$ pre-2020; ≈ 1.025 post-2020), indicating a slightly wider residual gap in the later period but no robust, incremental equity-like role for Litecoin after controls.

Table 69: Pre-2020 India Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{INR})$	-0.105*** (0.024)	-0.099*** (0.024)	-0.092*** (0.024)	-0.093*** (0.024)	-0.094*** (0.024)	-0.094*** (0.024)	-0.093*** (0.024)	-0.094*** (0.024)
$\Delta \text{Commodity}$		0.002*** (0.000)	0.001** (0.001)	0.001* (0.001)	0.001* (0.001)	0.001* (0.001)	0.001* (0.001)	0.001* (0.001)
ΔDollar			-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003** (0.001)
ΔVIX				-0.000** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
$\Delta 10\text{Y Yield}$					-0.009 (0.007)	-0.010 (0.007)	-0.009 (0.007)	-0.009 (0.007)
$\Delta 1\text{M Yield}$						0.006 (0.009)	0.006 (0.009)	0.006 (0.009)
$\Delta 10\text{Y Spread}$							-0.003 (0.006)	-0.001 (0.006)
$\Delta 2\text{Y Spread}$								-0.008 (0.007)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	0.020	0.034	0.042	0.048	0.048	0.048	0.047	0.047

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 70: Post-2020 India Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{INR})$	-1.578*** (0.079)	-1.549*** (0.079)	-1.316*** (0.081)	-1.309*** (0.080)	-1.285*** (0.080)	-1.285*** (0.080)	-1.280*** (0.081)	-1.273*** (0.081)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
ΔDollar			-0.006*** (0.001)	-0.005*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.011*** (0.004)	0.011*** (0.004)	0.011*** (0.004)	0.011*** (0.004)
$\Delta 1\text{M Yield}$						0.002 (0.004)	0.002 (0.004)	0.002 (0.004)
$\Delta 10\text{Y Spread}$							-0.002 (0.007)	0.002 (0.007)
$\Delta 2\text{Y Spread}$								-0.008 (0.005)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.159	0.166	0.201	0.231	0.234	0.234	0.234	0.234

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Interpretation. For India, the pre-2020 Tether results show a clear, modestly negative loading that survives controls: the USDT slope is -0.105 in the bivariate regression (col. 1) and remains around -0.093 to -0.094 once common-shock controls are added (cols. 4–8), while fit rises from $\text{Adj. } R^2 = 0.020$ to 0.047 . Post-2020, the negative comovement is far larger in the bivariate specification (-1.578 , col. 1) and, although

attenuated, it remains strongly negative with full controls (≈ -1.28 in cols. 7–8), with Adj. R^2 improving from 0.159 to ~ 0.234 . Before 2020, after accounting for broad macro factors (commodities, dollar, volatility, rates, term structure), $\Delta \log(\text{USD}\text{T}_{INR})$ retains a small but persistent inverse relation to equities; after 2020, the currency–equity linkage is much more pronounced, yielding economically large negative loadings even after controls. Using the square-root variance ratio, both windows lie *away* from unity ($\sqrt{\rho} \approx 1.299$ pre-2020; ≈ 4.339 post-2020), indicating a substantially wider residual gap in the later period, in line with stronger segmentation/flow effects and the markedly larger post-2020 coefficients.

Table 71: Pre-2020 India Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{INR})$	-0.008 (0.006)	-0.008 (0.006)	-0.008 (0.005)	-0.009 (0.005)	-0.009 (0.005)	-0.009 (0.005)	-0.009 (0.005)	-0.009 (0.005)
$\Delta \text{Commodity}$		0.002*** (0.001)	0.001 (0.001)	0.000 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
ΔDollar			-0.005*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
ΔVIX				-0.001** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					-0.013 (0.009)	-0.014 (0.009)	-0.013 (0.009)	-0.012 (0.009)
$\Delta 1\text{M Yield}$						0.008 (0.011)	0.008 (0.011)	0.008 (0.011)
$\Delta 10\text{Y Spread}$							-0.006 (0.007)	-0.003 (0.007)
$\Delta 2\text{Y Spread}$								-0.009 (0.008)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	0.002	0.014	0.031	0.037	0.039	0.038	0.038	0.038

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 72: Post-2020 India Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{INR})$	0.004 (0.004)	0.002 (0.004)	-0.001 (0.004)	-0.008** (0.004)	-0.008** (0.004)	-0.008** (0.004)	-0.008** (0.004)	-0.008** (0.004)
$\Delta \text{Commodity}$		0.001*** (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
ΔDollar			-0.009*** (0.001)	-0.008*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.017*** (0.004)	0.017*** (0.004)	0.018*** (0.004)	0.018*** (0.004)
$\Delta 1\text{M Yield}$						0.003 (0.004)	0.003 (0.004)	0.003 (0.004)
$\Delta 10\text{Y Spread}$							-0.019*** (0.007)	-0.012 (0.007)
$\Delta 2\text{Y Spread}$								-0.012** (0.005)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	-0.000	0.013	0.100	0.134	0.141	0.141	0.143	0.145

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For India, the pre-2020 regressions show weak, slightly negative comovement: the XRP slope is small in the bivariate specification (-0.008 , col. 1) and remains about -0.009 once common-shock controls are added (cols. 4–8), while fit rises (Adj. R^2 : $0.002 \rightarrow 0.038$). This is consistent with a pre-2020 “hedge/low comovement” pattern in which broad macro factors (commodities, dollar, volatility, rates, term structure) explain most equity variation, leaving $\Delta \log(\text{XRP}_{INR})$ with little incremental power. Post-2020, the bivariate XRP coefficient is small (0.004 , col. 1) but becomes clearly negative with controls (about -0.008 in cols. 4–8), even as Adj. R^2 improves from 0.000 to ~ 0.145 . Taken together, any raw comovement in the later period is largely absorbed by common shocks, leaving little robust equity-like loading for $\Delta \log(\text{XRP}_{INR})$. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie close to one ($\sqrt{\rho} \approx 1.019$ pre-2020; ≈ 1.023 post-2020), indicating similar residual parity across periods.

2.8 South Africa

Table 73: Pre-2020 South Africa Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{ZAR})$	-0.000 (0.002)	0.000 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.001 (0.002)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
ΔDollar			-0.006*** (0.001)	-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.000)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.002*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.020*** (0.003)	0.020*** (0.003)	0.020*** (0.003)	0.020*** (0.003)
$\Delta 1\text{M Yield}$						0.004 (0.006)	0.004 (0.006)	0.004 (0.006)
$\Delta 10\text{Y Spread}$							-0.000 (0.002)	-0.000 (0.002)
$\Delta 1\text{Y Spread}$								-0.001 (0.001)
Observations	3451	3451	3451	3451	3451	3451	3451	3451
Adj. R^2	-0.000	0.101	0.134	0.192	0.202	0.202	0.202	0.201

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For South Africa, the pre-2020 regressions show that as common-shock controls are added, the Bitcoin slope remains essentially nil in the bivariate specification (≈ 0.000 , col. 1) and stays economically tiny after controls (about 0.002 in cols. 4–6, easing to ≈ 0.001 with term spreads in cols. 7–8), while fit rises (Adj. R^2 : $-0.000 \rightarrow 0.201$). This is consistent with a pre-2020 “hedge/low comovement” narrative in which broad macro factors (commodities, dollar, volatility, rates, term structure) explain most equity variation, leaving $\Delta \log(\text{BTC}_{ZAR})$ with little incremental power. Post-2020, by contrast, the BTC coefficient is large and highly significant in the bivariate specification (0.053 , col. 1) and—though attenuated by controls—remains economically meaningful (≈ 0.016 in cols. 4–6, rebounding to ≈ 0.021 with term spreads in cols. 7–8), with Adj. R^2 improving from 0.025 to ~ 0.371 . This aligns with stronger crypto–FX–equity comovement after 2020, much of which is absorbed by common shocks but not fully eliminated. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.006$ pre-2020; ≈ 1.015 post-2020), indicating a slightly wider residual gap in the later window and reinforcing the shift from hedge-like behavior toward modest, persistent positive co-movement.

Interpretation. For South Africa, the pre-2020 regressions show that as common-shock controls are added, the Ethereum slope is small in the bivariate specification (0.005 , col. 1) and attenuates to a modest ≈ 0.003

Table 74: Post-2020 South Africa Bitcoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{ZAR})$	0.053*** (0.007)	0.042*** (0.007)	0.036*** (0.006)	0.017*** (0.006)	0.016*** (0.006)	0.016*** (0.006)	0.021*** (0.006)	0.021*** (0.006)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.012*** (0.001)	-0.011*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.020*** (0.004)	0.019*** (0.004)	0.023*** (0.004)	0.023*** (0.004)
$\Delta 1\text{M Yield}$						0.004 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta 10\text{Y Spread}$							-0.027*** (0.002)	-0.027*** (0.002)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.025	0.133	0.272	0.321	0.330	0.330	0.371	0.371

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 75: Pre-2020 South Africa Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{ZAR})$	0.005 (0.004)	0.005 (0.004)	0.005 (0.004)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
ΔDollar			-0.004*** (0.001)	-0.003*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.019*** (0.007)	0.019*** (0.007)	0.019*** (0.007)	0.019*** (0.007)
$\Delta 1\text{M Yield}$						0.008 (0.008)	0.008 (0.008)	0.008 (0.008)
$\Delta 10\text{Y Spread}$							0.007 (0.005)	0.007 (0.005)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	1092	1092	1092	1092	1092	1092	1092	1092
Adj. R^2	0.001	0.075	0.091	0.136	0.141	0.141	0.142	0.142

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 76: Post-2020 South Africa Ethereum Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{ZAR})$	0.047*** (0.005)	0.038*** (0.005)	0.030*** (0.005)	0.014*** (0.005)	0.014*** (0.005)	0.014*** (0.005)	0.016*** (0.005)	0.016*** (0.005)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.012*** (0.001)	-0.011*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.009*** (0.001)	-0.009*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10Y \text{ Yield}$					0.020*** (0.004)	0.019*** (0.004)	0.023*** (0.004)	0.023*** (0.004)
$\Delta 1M \text{ Yield}$						0.004 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta 10Y \text{ Spread}$							-0.027*** (0.002)	-0.027*** (0.002)
$\Delta 1Y \text{ Spread}$								0.000 (0.000)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.035	0.139	0.275	0.321	0.331	0.331	0.370	0.370

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

once controls are included (cols. 4–8), while fit rises (Adj. R^2 : $0.001 \rightarrow 0.142$). This is consistent with a pre-2020 “hedge/low comovement” narrative in which broad macro factors (commodities, dollar, volatility, rates, term structure) account for most equity variation, leaving $\Delta \log(\text{ETH}_{ZAR})$ with limited incremental power. Post-2020, by contrast, the ETH coefficient is large and highly significant in the bivariate specification (0.047, col. 1) and—though reduced by controls—remains economically meaningful (≈ 0.014 in cols. 4–6 and ≈ 0.016 with term spreads in cols. 7–8), with Adj. R^2 improving from 0.035 to ~ 0.370 . This aligns with stronger crypto–FX–equity comovement after 2020, much of which is absorbed by common shocks but not eliminated. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie near unity ($\sqrt{\rho} \approx 1.003$ pre-2020; ≈ 1.006 post-2020), indicating a slightly wider residual gap in the later window and reinforcing the shift from hedge-like behavior toward sustained positive co-movement.

Interpretation. For South Africa, the pre-2020 regressions show weak, imprecise comovement: the Litecoin slope is near zero in the bivariate specification (-0.003 , col. 1) and remains economically small once common-shock controls are added (settling around ≈ 0.001 in cols. 4–8), while fit rises (Adj. R^2 : $-0.001 \rightarrow 0.158$). This is consistent with a pre-2020 “hedge/low comovement” pattern in which broad macro factors (commodities, dollar, volatility, rates, term structure) explain most equity variation, leaving $\Delta \log(\text{LTC}_{ZAR})$ with little incremental power. Post-2020, the bivariate LTC coefficient is clearly positive and significant (0.029, col. 1) but attenuates once controls are included (to about 0.007 in cols. 4–6 and ≈ 0.010 in cols. 7–8), even as Adj. R^2 improves from 0.017 to ~ 0.368 . Taken together, the later-period comovement strengthens but is largely absorbed by common shocks, leaving a modest residual role for $\Delta \log(\text{LTC}_{ZAR})$ after controls. Using the square-root variance ratio $\sqrt{\rho}$, both windows lie close to one ($\sqrt{\rho} \approx 1.009$ pre-2020; ≈ 1.007 post-2020), implying only a slightly looser parity in the early window and supporting the narrative of stronger—but mostly common-shock—driven equity-like loading after 2020.

Interpretation. For South Africa, the pre-2020 regressions show a modest, mostly negative bivariate loading of Tether on equities (-0.102 , col. 1) that is quickly absorbed by common shocks: once controls enter, the USDT coefficient shrinks toward zero and becomes statistically weak (cols. 3–8), while fit rises from 0.025 to ~ 0.158 (Adj. R^2). This is consistent with a pre-2020 “hedge/low comovement” pattern in which broad macro factors (commodities, dollar, volatility, rates, term structure) explain most equity variation, leaving $\Delta \log(\text{USDT}_{ZAR})$ with little incremental power. Post-2020, by contrast, the bivariate effect is large and negative (-0.421 , col. 1) but attenuates and even flips sign after controls, settling near

Table 77: Pre-2020 South Africa Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{ZAR})$	-0.003 (0.006)	-0.001 (0.006)	0.001 (0.006)	0.001 (0.005)	0.001 (0.005)	0.001 (0.005)	0.001 (0.005)	0.001 (0.005)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.003*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.008*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.015 (0.009)	0.014 (0.009)	0.013 (0.009)	0.013 (0.009)
$\Delta 1\text{M Yield}$						0.012 (0.012)	0.011 (0.012)	0.011 (0.012)
$\Delta 10\text{Y Spread}$							0.009 (0.007)	0.009 (0.007)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	-0.001	0.080	0.120	0.156	0.158	0.158	0.158	0.158

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 78: Post-2020 South Africa Litecoin Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{ZAR})$	0.029*** (0.005)	0.023*** (0.005)	0.020*** (0.004)	0.007* (0.004)	0.007* (0.004)	0.007* (0.004)	0.010** (0.004)	0.010** (0.004)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.012*** (0.001)	-0.011*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.020*** (0.004)	0.020*** (0.004)	0.023*** (0.004)	0.023*** (0.004)
$\Delta 1\text{M Yield}$						0.004 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta 10\text{Y Spread}$							-0.027*** (0.002)	-0.027*** (0.002)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.017	0.127	0.268	0.319	0.329	0.329	0.368	0.368

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 79: Pre-2020 South Africa Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{ZAR})$	-0.102*** (0.021)	-0.065*** (0.021)	-0.032 (0.022)	-0.015 (0.021)	-0.015 (0.021)	-0.015 (0.021)	-0.020 (0.022)	-0.020 (0.022)
$\Delta \text{Commodity}$		0.004*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)
ΔDollar			-0.006*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.014* (0.008)	0.014* (0.008)	0.013* (0.008)	0.013* (0.008)
$\Delta 1\text{M Yield}$						0.005 (0.010)	0.005 (0.010)	0.005 (0.010)
$\Delta 10\text{Y Spread}$							0.009 (0.006)	0.009 (0.006)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	879	879	879	879	879	879	879	879
Adj. R^2	0.025	0.091	0.113	0.156	0.158	0.157	0.158	0.158

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 80: Post-2020 South Africa Tether Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{ZAR})$	-0.421*** (0.028)	-0.322*** (0.028)	-0.084*** (0.030)	0.023 (0.030)	0.011 (0.030)	0.010 (0.030)	0.107*** (0.030)	0.107*** (0.030)
$\Delta \text{Commodity}$		0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.011*** (0.001)	-0.011*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.020*** (0.004)	0.019*** (0.004)	0.023*** (0.004)	0.023*** (0.004)
$\Delta 1\text{M Yield}$						0.004 (0.003)	0.002 (0.003)	0.002 (0.003)
$\Delta 10\text{Y Spread}$							-0.029*** (0.002)	-0.029*** (0.002)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.099	0.170	0.263	0.319	0.328	0.328	0.370	0.370

Note: Dependent variable shown in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

a small positive and significant loading (≈ 0.107 in cols. 7–8), as Adj. R^2 improves from 0.099 to ~ 0.370 . Using the square-root variance ratio $\sqrt{\rho}$, parity loosens from $\sqrt{\rho} \approx 1.204$ pre-2020 to ≈ 1.630 post-2020, indicating a larger residual gap in the later window. Taken together, raw comovement strengthens after 2020 but is largely mediated by common shocks, with a residual—now slightly positive—USDT–equity linkage emerging only once the macro channels are netted out.

Table 81: Pre-2020 South Africa Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{ZAR})$	0.000 (0.006)	0.003 (0.006)	0.005 (0.006)	0.005 (0.006)	0.005 (0.006)	0.005 (0.006)	0.005 (0.006)	0.005 (0.006)
$\Delta \text{Commodity}$		0.005*** (0.001)	0.003*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)
ΔDollar			-0.008*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.015 (0.009)	0.014 (0.009)	0.013 (0.009)	0.013 (0.009)
$\Delta 1\text{M Yield}$						0.012 (0.012)	0.011 (0.012)	0.011 (0.012)
$\Delta 10\text{Y Spread}$							0.008 (0.007)	0.008 (0.007)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	691	691	691	691	691	691	691	691
Adj. R^2	-0.001	0.081	0.121	0.157	0.158	0.158	0.159	0.159

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 82: Post-2020 South Africa Ripple Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{ZAR})$	0.016*** (0.004)	0.012*** (0.004)	0.012*** (0.004)	0.005 (0.004)	0.005 (0.004)	0.005 (0.004)	0.007** (0.003)	0.007** (0.003)
$\Delta \text{Commodity}$		0.004*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
ΔDollar			-0.012*** (0.001)	-0.011*** (0.001)	-0.012*** (0.001)	-0.012*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)
ΔVIX				-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
$\Delta 10\text{Y Yield}$					0.020*** (0.004)	0.020*** (0.004)	0.023*** (0.004)	0.023*** (0.004)
$\Delta 1\text{M Yield}$						0.004 (0.003)	0.003 (0.003)	0.003 (0.003)
$\Delta 10\text{Y Spread}$							-0.027*** (0.002)	-0.027*** (0.002)
$\Delta 1\text{Y Spread}$								0.000 (0.000)
Observations	2087	2087	2087	2087	2087	2087	2087	2087
Adj. R^2	0.006	0.120	0.264	0.319	0.328	0.329	0.368	0.368

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Interpretation. For South Africa, the pre-2020 regressions show weak comovement: the XRP slope is essentially zero in the bivariate specification (0.000, col. 1) and remains economically small once common-

shock controls are added (drifting to ≈ 0.005 by cols. 4–8), while fit rises (Adj. R^2 : $-0.001 \rightarrow 0.159$). This matches a pre-2020 “hedge/low comovement” pattern in which broad macro factors (commodities, dollar, volatility, rates, term structure) explain most equity variation, leaving $\Delta \log(\text{XRP}_{ZAR})$ with little incremental power. Post-2020, by contrast, the bivariate coefficient is larger and significant (0.016, col. 1) but attenuates once controls are included, settling near a small positive loading (≈ 0.005 in cols. 4–6 and ≈ 0.007 with term spreads in cols. 7–8), even as Adj. R^2 improves from 0.006 to ~ 0.368 . Using the square-root variance ratio $\sqrt{\rho}$, both windows lie very close to one ($\sqrt{\rho} \approx 1.007$ pre-2020; ≈ 1.005 post-2020), indicating a slightly tighter parity (smaller residual gap) in the later period. Overall, raw XRP–equity comovement strengthens after 2020 but is largely absorbed by common shocks, leaving only a modest, residual positive loading once those shocks are netted out.